CS1090A — Lecture 3: Visualization and EDA

CS109A Introduction to Data Science

Pavlos Protopapas, Kevin Rader and Chris Gumb



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Words for the same idea

- row = **observation** · **record** · **instance**
- column = **variable** · **attribute** · **feature** · **dimension**
- **sample** — **the observations you actually collected**, out of all the ones you could have (or a subset of the data generally)

one column = one variable

one row = one observation

In [4]:

```
imdb = pd.read_csv('imdb_top_1000.csv')
imdb.head()
```

Out[4]:

	Poster_Link	Series_Title	Released_Year	Certificate	Runtime	Genre	IMDB_Rating	Overview	Meta_score	Director	Star1
0	https://m.media-amazon.com/images/M/MV5BMDFkYT...	The Shawshank Redemption	1994	A	142 min	Drama	9.3	Two imprisoned men bond over a number of years...	80.0	Frank Darabont	Tim Robbins
1	https://m.media-amazon.com/images/M/MV5BM2MyNj...	The Godfather	1972	A	175 min	Crime, Drama	9.2	An organized crime dynasty's aging patriarch t...	100.0	Francis Ford Coppola	Marlon Brando
2	https://m.media-amazon.com/images/M/MV5BMTMxNT...	The Dark Knight	2008	UA	152 min	Action, Crime, Drama	9.0	When the menace known as the Joker wreaks havo...	84.0	Christopher Nolan	Christian Bale
3	https://m.media-amazon.com/images/M/MV5BMWMwMG...	The Godfather: Part II	1974	A	202 min	Crime, Drama	9.0	The early life and career of Vito Corleone	90.0	Francis Ford Coppola	Al Pacino

Messy Data

The following is a table accounting for the number of produce deliveries over a weekend.

What are the variables in this dataset? What object or event are we measuring?

	Friday	Saturday	Sunday
Morning	15	158	10
Afternoon	2	90	20
Evening	55	12	45

What's the issue? How do we fix it?

ID	Time	Day	Number
1	Morning	Friday	15
2	Morning	Saturday	158
3	Morning	Sunday	10
4	Afternoon	Friday	2
5	Afternoon	Saturday	9
6	Afternoon	Sunday	20
7	Evening	Friday	55
8	Evening	Saturday	12
9	Evening	Sunday	45

Tabular = Happy Pavlos



Common causes of messiness are:

- Column headers are values, not variable names
- Variables are stored in both rows and columns
- Multiple variables are stored in one column/entry
- Multiple types of experimental units stored in same table

In general, we want each file to correspond to a dataset, each column to represent a single variable and each row to represent a single observation.

We want to **tabularize** the data. This makes pandas, sklearn, statsmodels, etc. in python happy.

Lecture Outline: Data, Summaries, and Visuals

- **Exploratory Data Analysis (EDA)**
  - **Descriptive Statistics**
  - Basic Visualizations

- Historical Interlude

- Effective Visualizations

Reading: Ch. 1 in *An Introduction to Statistical Learning* (ISL)

Basics of Sampling

Population versus sample:

- A **population** is the entire set of objects or events under study. Population can be hypothetical “all students” or all students in this class.
- A **sample** is a “representative” subset of the objects or events under study. Needed because it’s impossible or intractable to obtain or compute with population data.

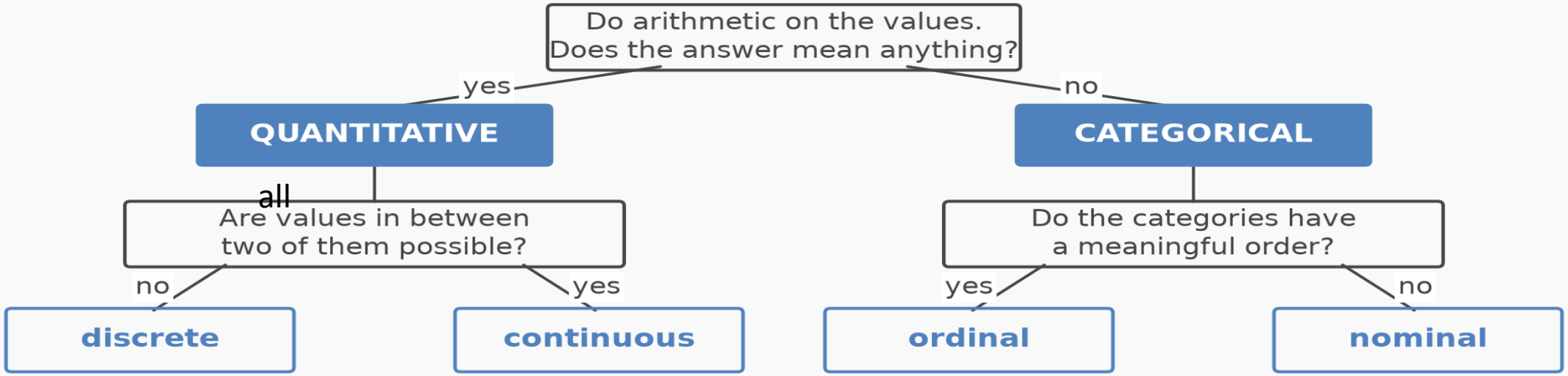
Biases in samples:

- **Selection bias**: some subjects or records are more likely to be selected
- Volunteer/**nonresponse bias**: subjects or records who are not easily available are not represented

Examples? Early adopters. Online responders. Students in class. Etc.

Asking "what kind of variable is this?" first

- First ask: **quantitative, ordinal, or nominal?**
- The type says **which summaries have a meaning**
- Ask ”what kind is this variable” before you try to compute anything!

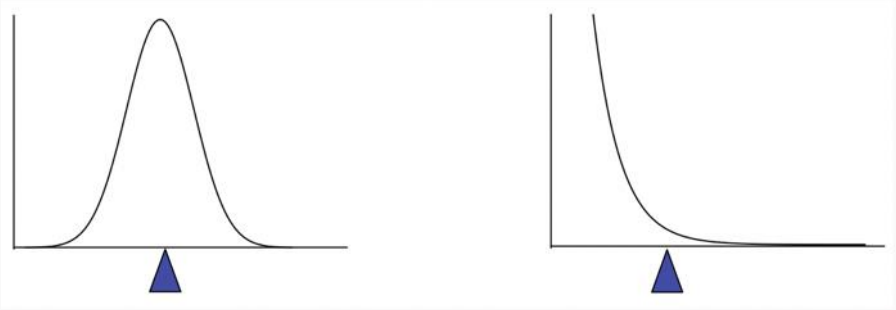


A postcode is a number and is nominal — the arithmetic runs, and the answer means nothing.

Sample mean

The **mean** of a set of *n* observations of a variable is denoted $\bar{x}$ and is defined as:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i$$



The mean describes what a “typical” sample value looks like, or where is the “center” of the distribution of the data.

Key theme: there is always uncertainty involved when calculating a sample mean to estimate a population mean.

Sample median

The **median** of a set of n number of observations in a sample, ordered by value, of a variable is is defined by

$$\text{Median} = \begin{cases} x_{(n+1)/2} & \text{if } n \text{ is odd} \\ \frac{x_{n/2} + x_{(n+1)/2}}{2} & \text{if } n \text{ is even} \end{cases}$$

Example (already in order):

Ages: 17, 19, 21, 22, 23, 23, 23, 38

Median = (22+23)/2 = 22.5

The median also describes what a typical observation looks like, or where is the center of the distribution of the sample of observations.

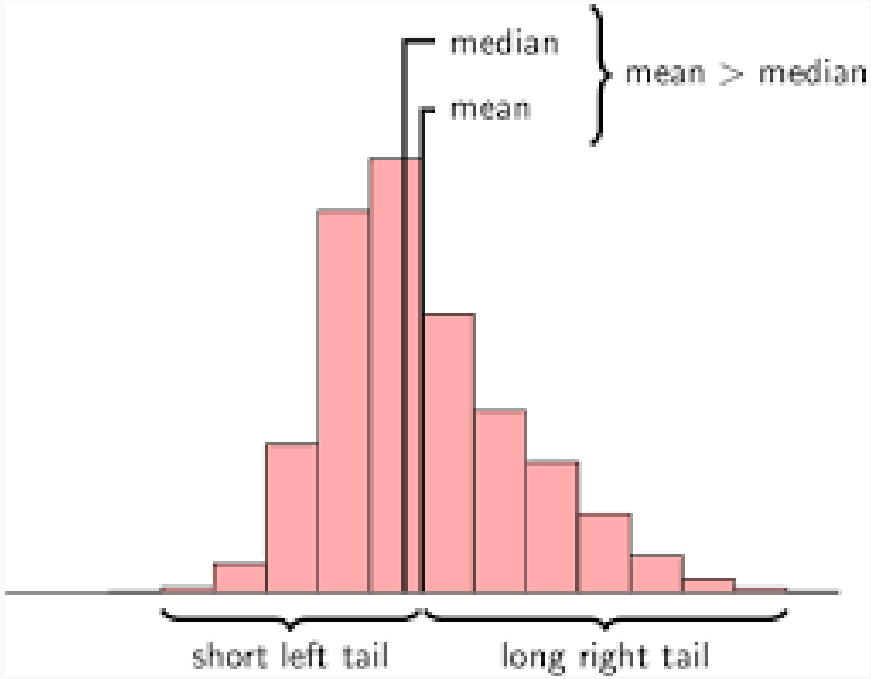
Mean vs. Median

The mean is sensitive to extreme values (**outliers**)



Mean, median, and skewness

The mean is sensitive to outliers:



The above distribution is called **right-skewed** since the mean is greater than the median. Note: **skewness** often "follows the longer tail".

Computational time

How hard (in terms of algorithmic complexity) is it to calculate

- the mean?

  at most $O(n)$

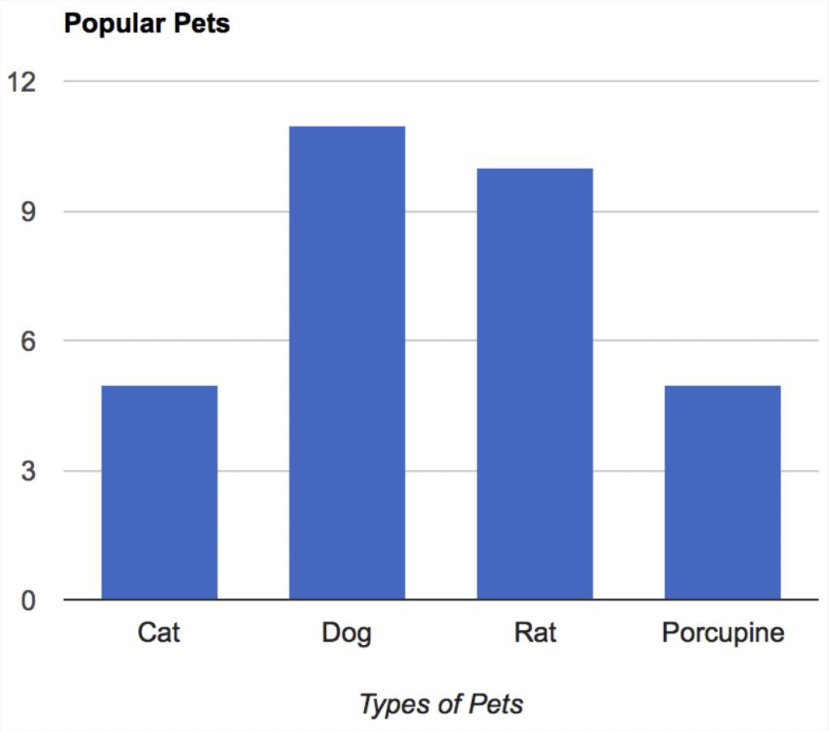
- the median?

  at most $O(n \log n)$ [or possibly $O(n)$ if you are lucky]

Note: Practicality of implementation should be considered!

Regarding Categorical Variables…

For categorical variables, neither mean or median make sense. Why?



The mode might be a better way to find the most "representative" value.

Measures of Spread: Range

The spread of a sample of observations measures how well the mean or median describes the sample.

One way to measure spread of a sample of observations is via the **range**.

Range = Maximum Value - Minimum Value

Measures of Spread: Variance

The (sample) **variance**, denoted s^2, measures how much on average the sample values deviate from the mean:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n} |x_i - \bar{x}|^2$$

Note: the term $|x_i - \bar{x}|$ measures the amount by which each x_i deviates from the mean $\bar{x}$. Squaring these deviations means that s^2 is sensitive to extreme values (outliers).

Note: s^2 doesn't have the same units as the x_i :(

What does a variance of 1,008 mean? Or 0.0001?

Measures of Spread: Standard Deviation

The (sample) **standard deviation**, denoted s, is the square root of the variance:

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n} |x_i - \bar{x}|^2}$$

Note: s does have the same units as the x_i. Phew!

Standard deviation is the "average" spread of the observations around the mean.

Measures of Association: Correlation Coefficient

The (sample) **correlation** between two variables X and Y measured on the same n observations is given by the formula:

$$r = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s_x}\right)\left(\frac{y_i - \bar{y}}{s_y}\right)$$

Note: r is unitless and ranges from -1 to +1

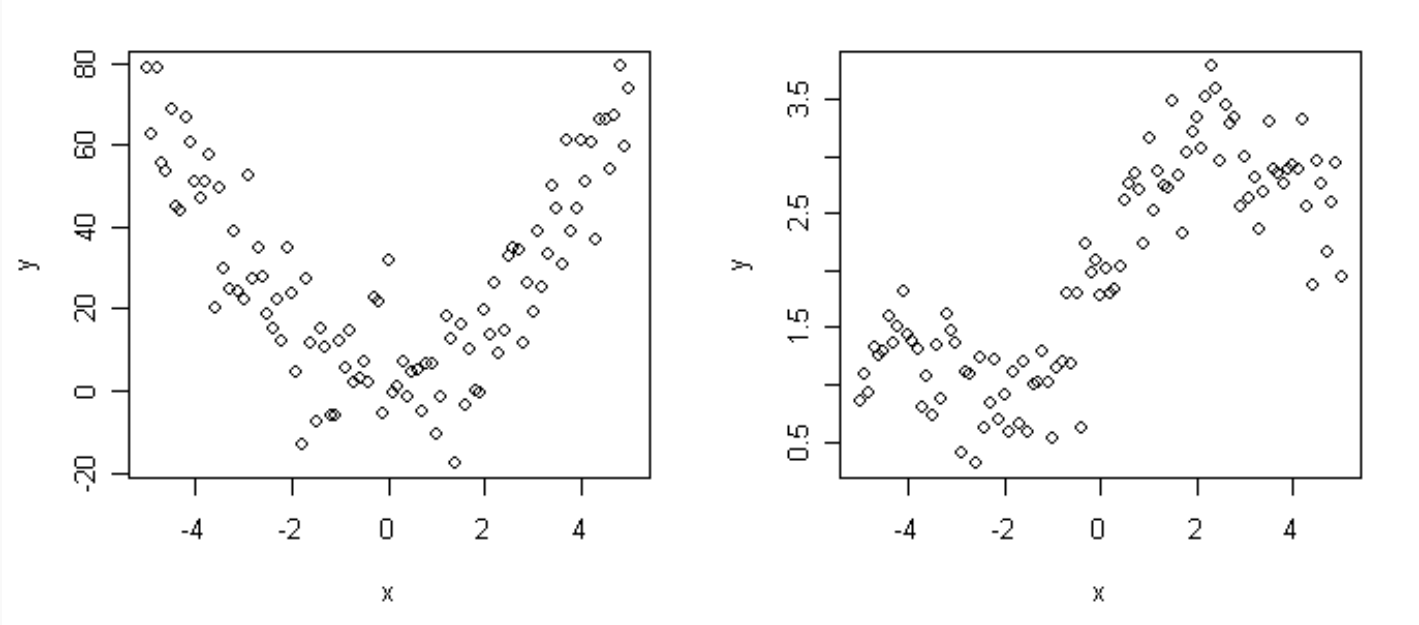
Correlation measures how to numeric variables are linearly related.

Keep in mind: correlation does not always imply causation.

Cautions about Correlation Coefficient

Always plot your data! Correlation does not do a good job of characterizing the strength of a *nonlinear* relationship:

- A high correlation does not always imply a perfect linear relationship
- A low correlation does not always imply no relationship.



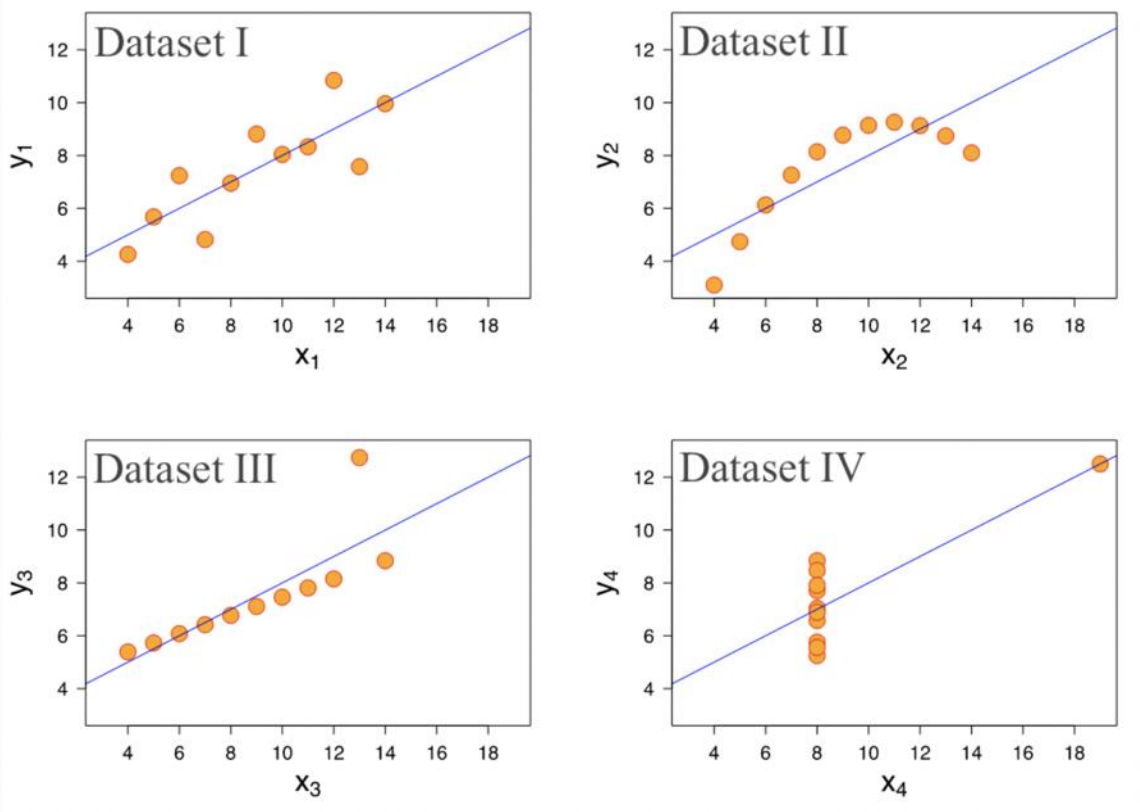
Anscombe's Data

The following four data sets comprise the Anscombe's Quartet; all four sets of data have identical simple summary statistics.

	Dataset I		Dataset II		Dataset III		Dataset IV	
	x	y	x	y	x	y	x	y
	10	8.04	10	9.14	10	7.46	8	6.58
	8	6.95	8	8.14	8	6.77	8	5.76
	13	7.58	13	8.74	13	12.74	8	7.71
	9	8.81	9	8.77	9	7.11	8	8.84
	11	8.33	11	9.26	11	7.81	8	8.47
	14	9.96	14	8.1	14	8.84	8	7.04
	6	7.24	6	6.13	6	6.08	8	5.25
	4	4.26	4	3.1	4	5.39	19	12.5
	12	10.84	12	9.13	12	8.15	8	5.56
	7	4.82	7	7.26	7	6.42	8	7.91
	5	5.68	5	4.74	5	5.73	8	6.89
Sum:	99.00	82.51	99.00	82.51	99.00	82.51	99.00	82.51
Avg:	9.00	7.50	9.00	7.50	9.00	7.50	9.00	7.50
Std:	3.32	2.03	3.32	2.03	3.32	2.03	3.32	2.03

Anscombe's Data (cont.)

Summary statistics clearly don't tell the story of how they differ. But a picture can be worth a thousand words:



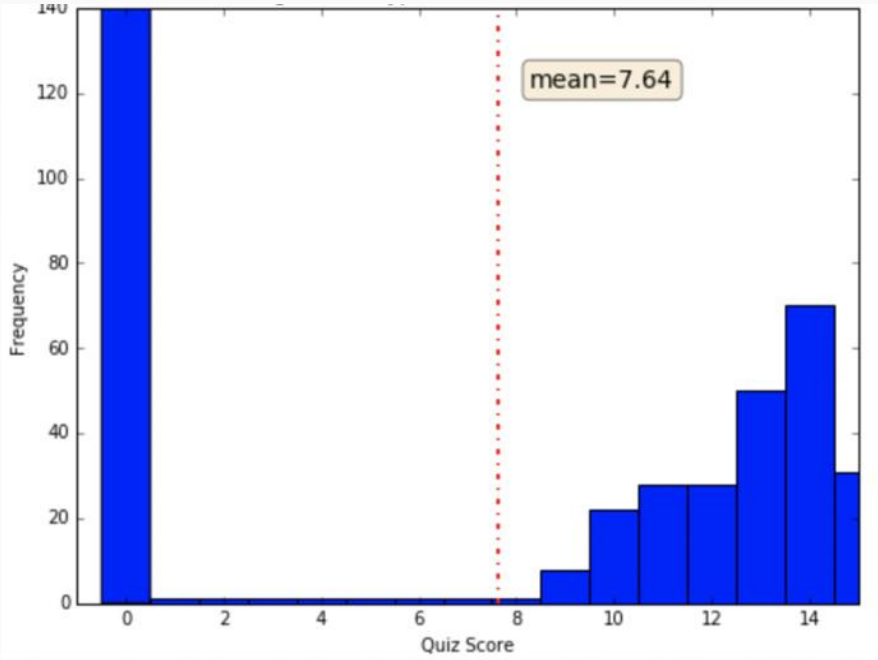
Lecture Outline: Data, Summaries, and Visuals

- What are Data?
- **Exploratory Data Analysis (EDA)**
  - Descriptive Statistics
  - **Basic Visualizations**
- Historical Interlude
- Effective Visualizations

Reading: Ch. 1 in *An Introduction to Statistical Learning* (ISL)

More Visualization Motivation

If I tell you that the average score for Homework 0 in my other class was: 7.64/15 = 50.9%, what does that suggest?



And what does the graph suggest?

More Visualization Motivation

Visualizations help us to analyze and explore the data. They help to:

- Identify hidden patterns and trends
- Formulate/test hypotheses
- Communicate any modeling results
  - Present information and ideas succinctly
  - Provide evidence and support
  - Influence and persuade
- Determine the next step in analysis/modeling

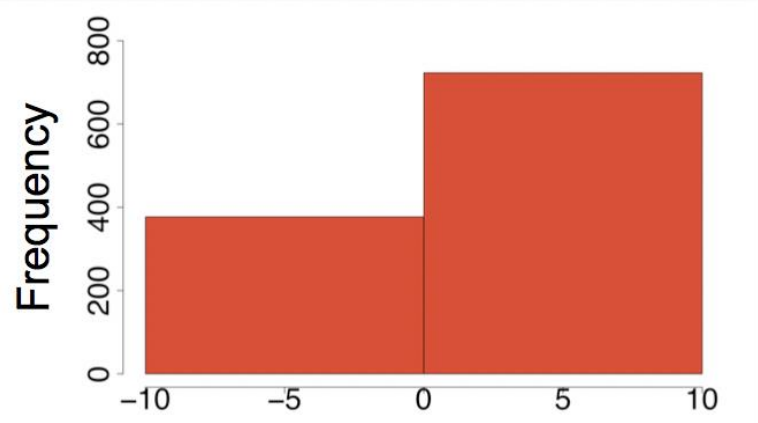
Types of Visualizations

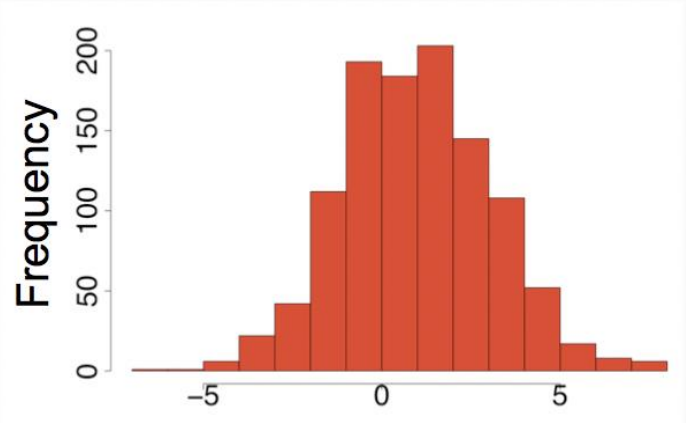
What do you want your visualization to show about your data?

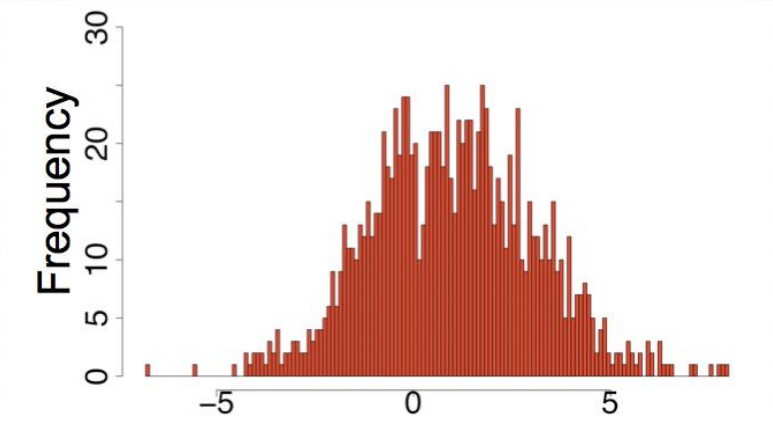
- **Distribution:** how a variable or variables in the dataset distribute over a range of possible values.
- **Relationship:** how the values of multiple variables in the dataset relate
- **Composition:** how the dataset breaks down into subgroups
- **Comparison:** how trends in multiple variable or datasets compare

Histograms to visualize distribution

A **histogram** is a way to visualize how 1-dimensional numeric data is distributed across certain values.



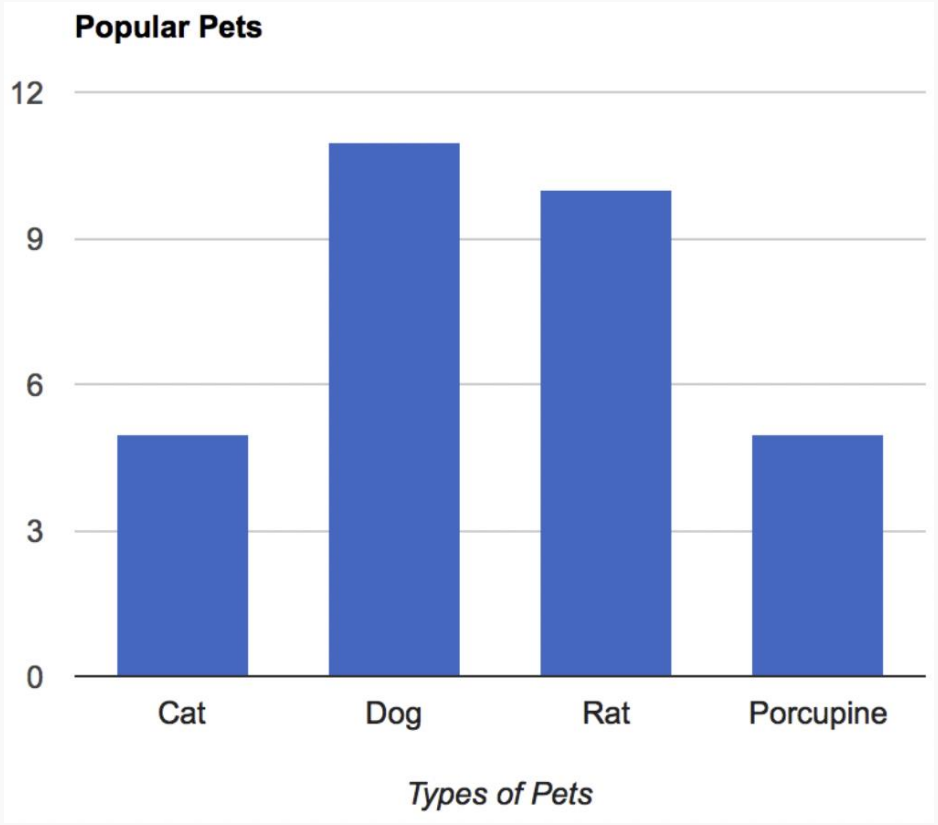




Note: Patterns in histograms are sensitive to the number of bins.

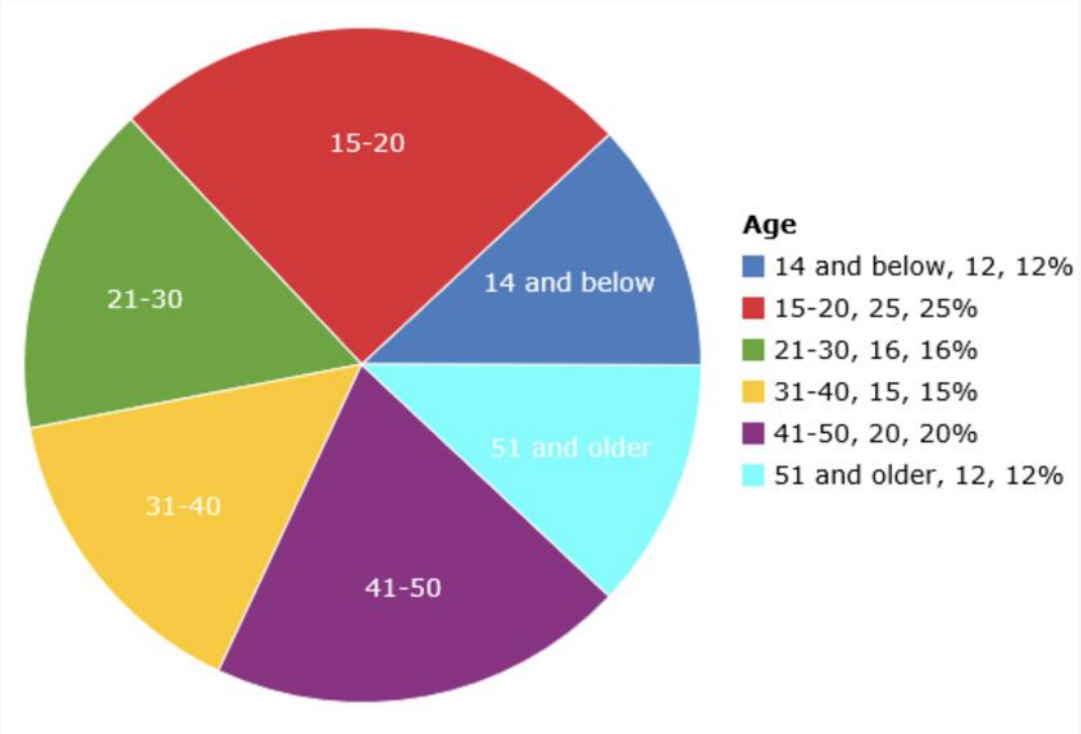
Bar plot for a categorical variable

A **bar plot** is a way to visualize the distribution of a categorical variable.



Pie chart for a categorical variable?

A **pie chart** is a way to visualize the static composition (aka, distribution) of a variable (or single group).



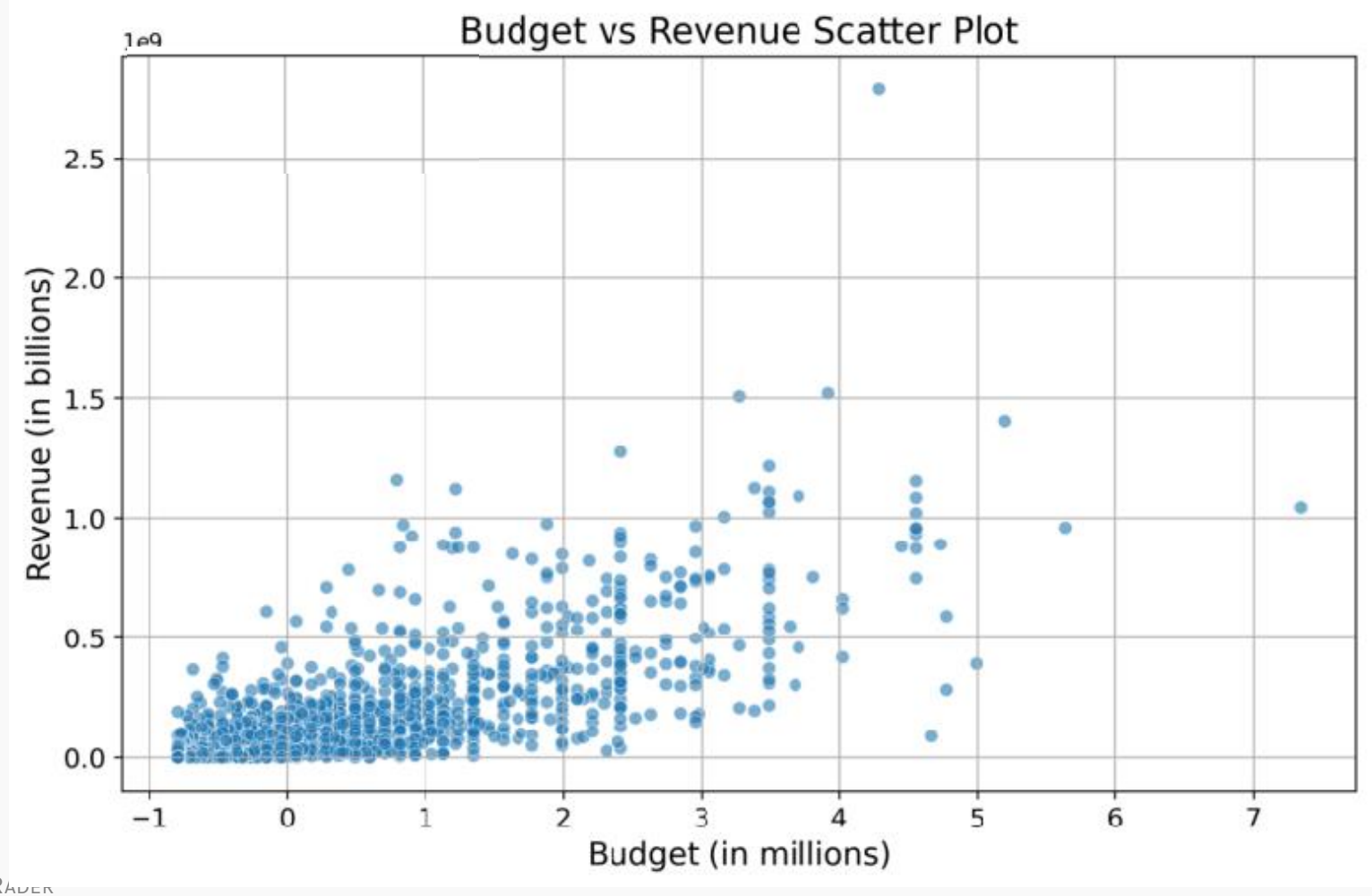
Pie charts are often frowned upon (and bar charts are used instead). Why?

Pies vs. Bars


MY FAVORITE BARS
(In Percentage of Awesomeness)
McGee's
MacLaren's
28%
30%
12%
27%
White Horse Tavern
3%
King Cole Bar
P & G
MY FAVORITE PIES
(In Percentage of Tastiness)

Scatter plots to visualize relationships

A **scatter plot** is a way to visualize the relationship between two different attributes of multi-dimensional data.



$r = \underline{0.772}$

How would you describe this scatterplot? What do you notice? What next *data wrangling* steps would you consider

Scatter plots to visualize relationships

What does ChatGPT think?

describe the relationship between the variables in the scatterplot:

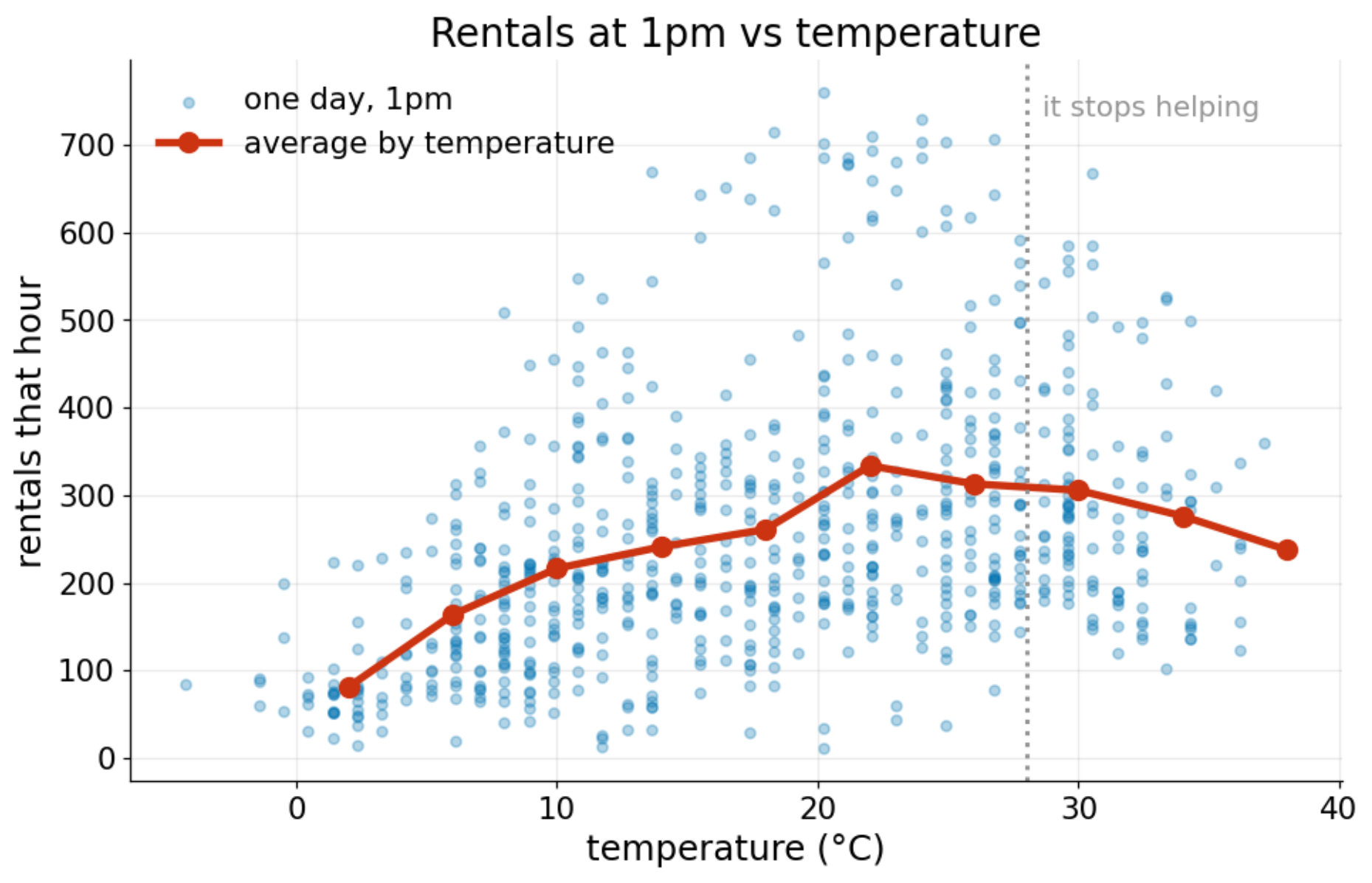
The scatterplot shows the relationship between **movie budget** and **box-office revenue**. According to the figure's accompanying text, the correlation coefficient is **0.77**. ResearchGate

- **Direction:** The relationship is **positive**. As a movie's budget increases, its box-office revenue generally tends to increase as well.
- **Strength:** A correlation of **0.77** indicates a **fairly strong positive linear relationship**. The points tend to follow an upward pattern rather than being randomly scattered.
- **Interpretation:** Movies with larger production budgets tend, on average, to earn more at the box office.
- **Variation:** The relationship is not perfect—there are still movies that earn unusually high or low revenue relative to their budgets. This means budget alone does not determine box-office success.
- **Important caution:** The correlation does **not** prove that a higher budget *causes* higher box-office revenue. Other factors—such as marketing, cast, genre, release timing, and audience interest—could also influence revenue.

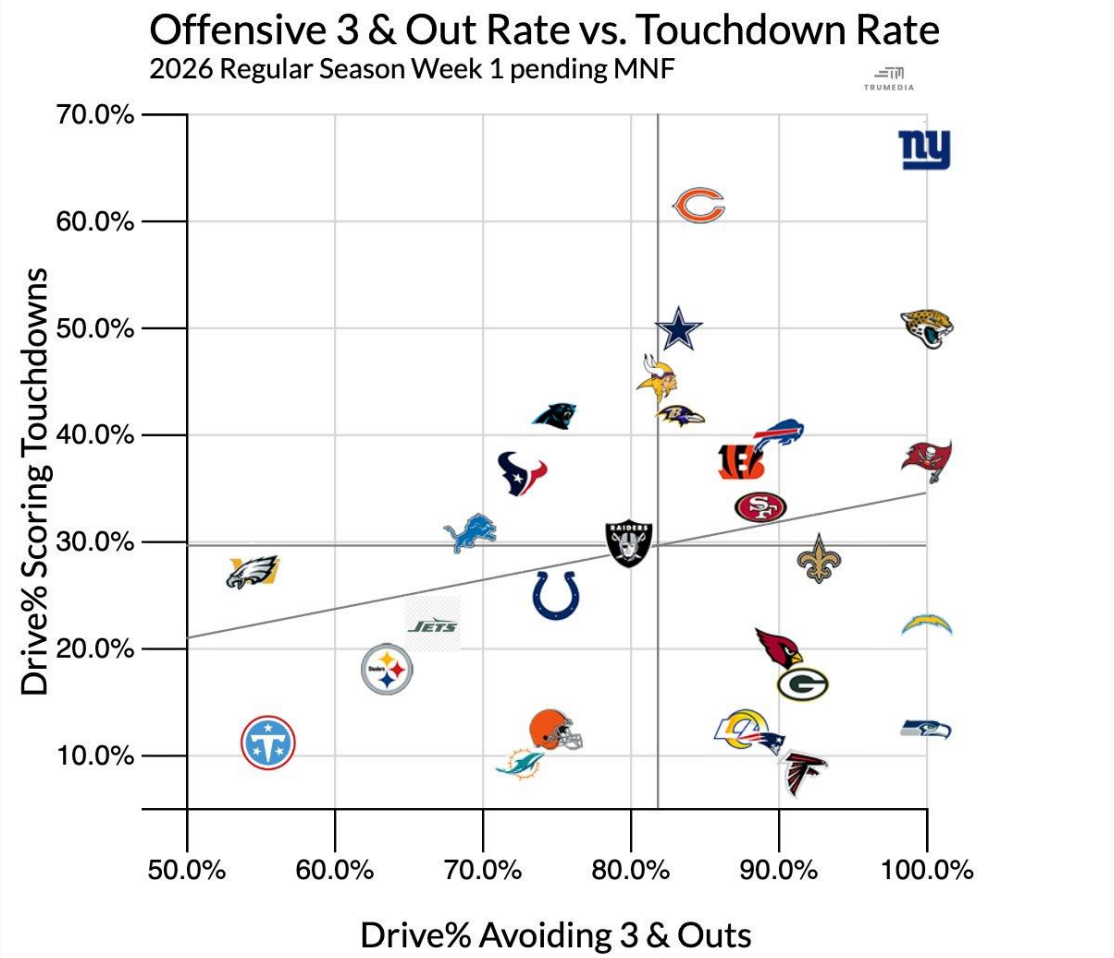
In one sentence: The scatterplot indicates a **strong positive association between movie budget and box-office revenue (r = 0.77)**, meaning higher-budget movies generally tend to generate higher box-office earnings. ResearchGate

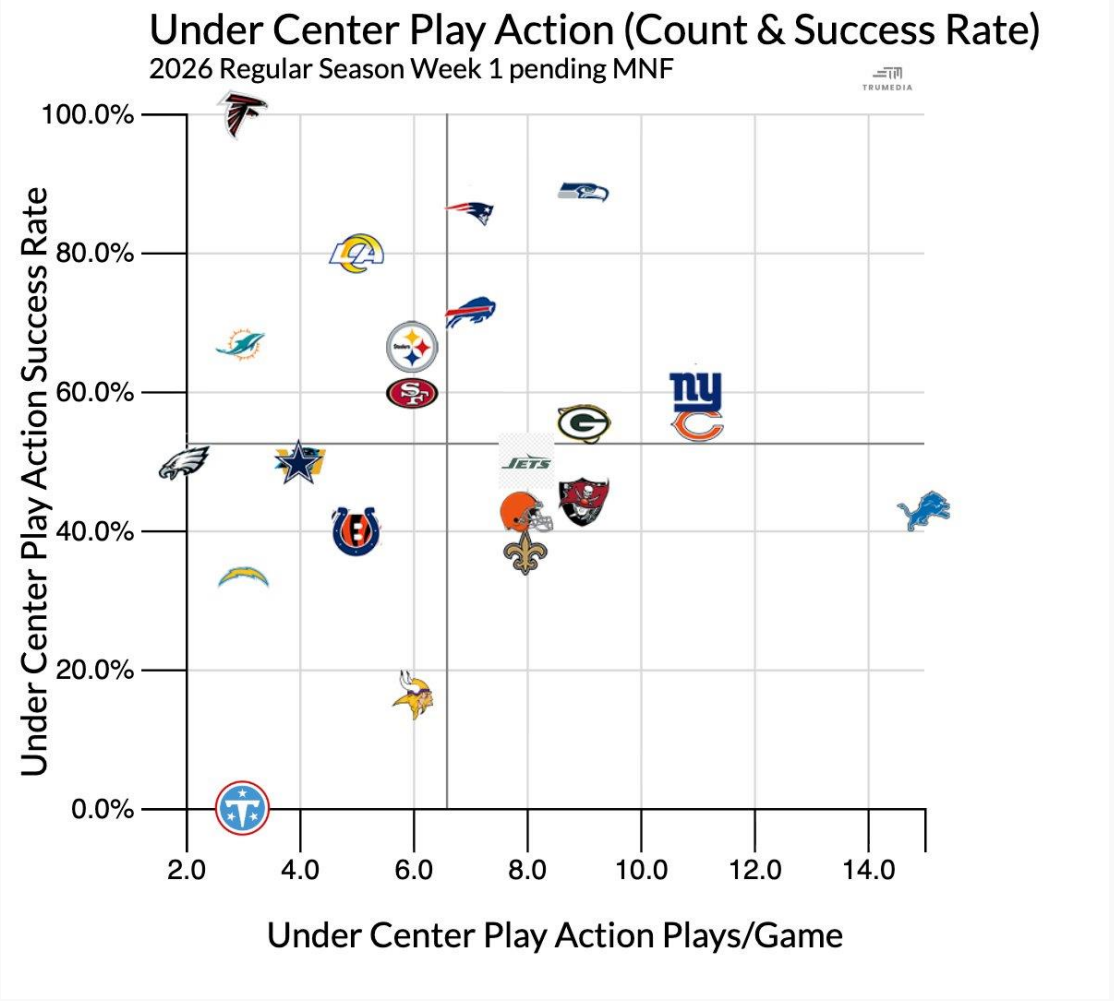
What did ChatGPT miss? What could we do better?

From the first lecture:



Scatter plots to visualize relationships

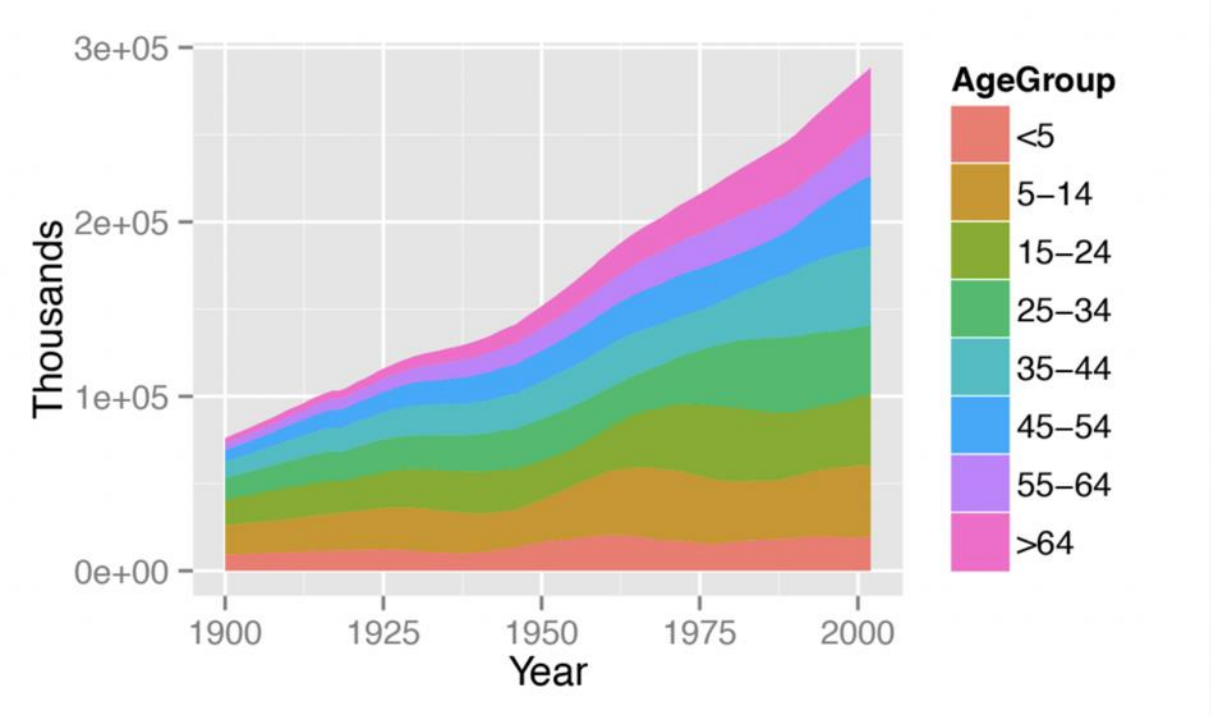




Thanks to [Deniz Selman]

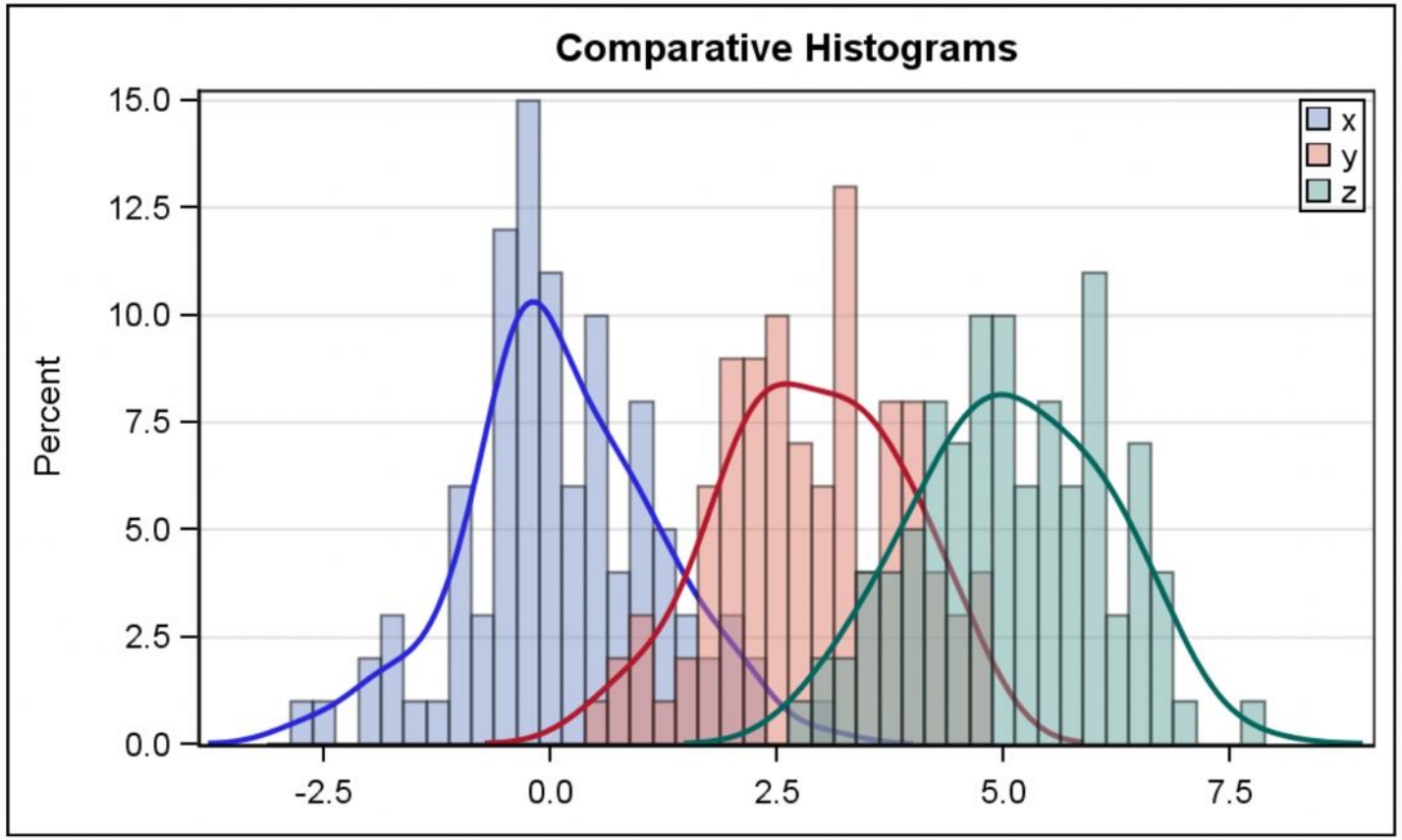
Stacked area graph to show trend over time

A **stacked area graph** is a way to visualize the composition of a group as it changes over time (or some other quantitative variable). This shows the relationship of a categorical variable (AgeGroup) to a quantitative variable (year).



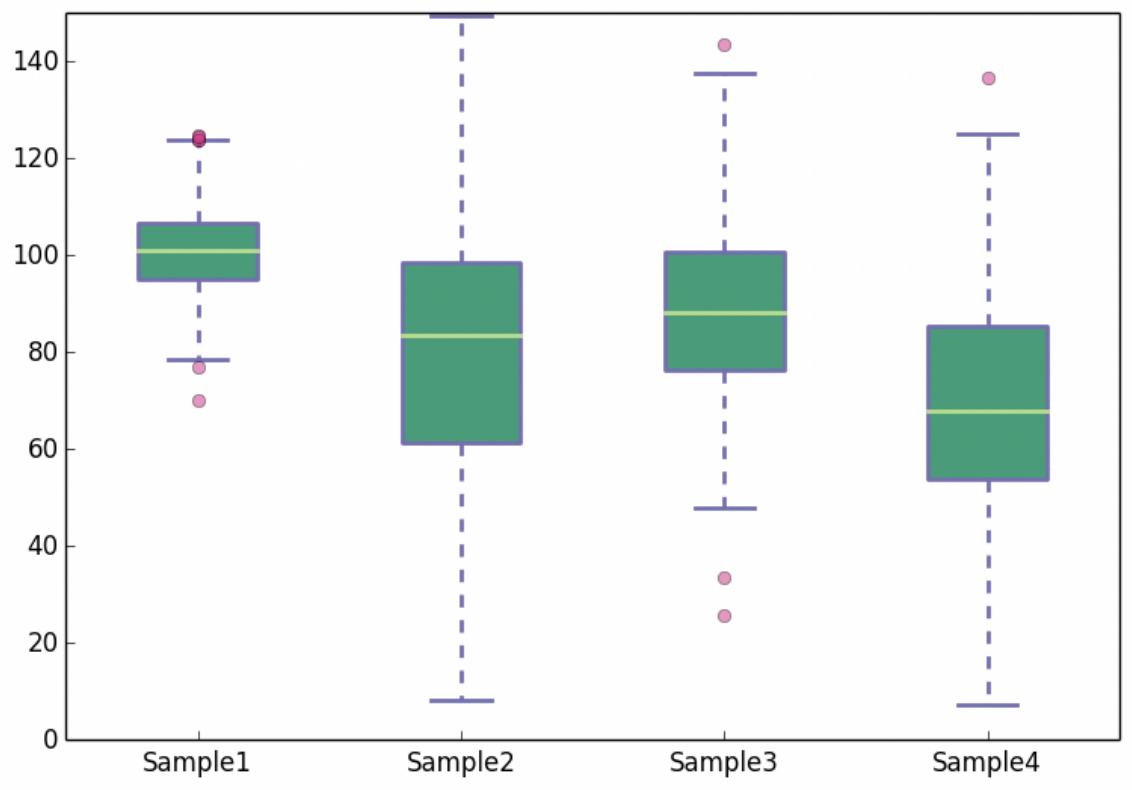
Multiple histograms

Plotting **multiple histograms (**and **kernel density estimates** of the distribution, here) on the same axes is a way to visualize how different variables compare (or how a variable differs over specific groups).



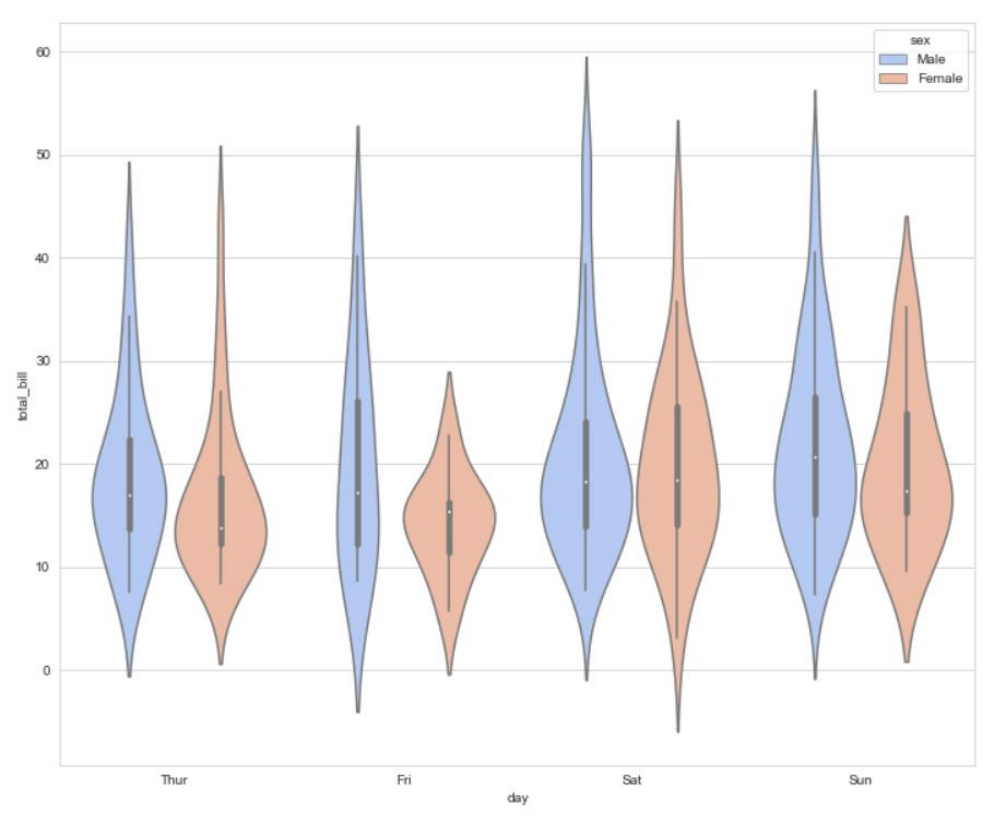
Boxplots

A **boxplot** is a simplified visualization to compare a quantitative variable across groups. It highlights the range, quartiles, median and any outliers present in a data set.



Violin plots

A **violin plot** is useful to show a little more detail than the boxplot. It mirrors 2 versions of the kernel density estimate (KDE) and puts them back-to-back.



[Not] Anything is possible!

Often your dataset seem too complex to visualize:

- Data is too high dimensional (how do you plot 100 variables on the same set of axes?)
- Some variables are categorical (how do you plot values like Yes or No?)

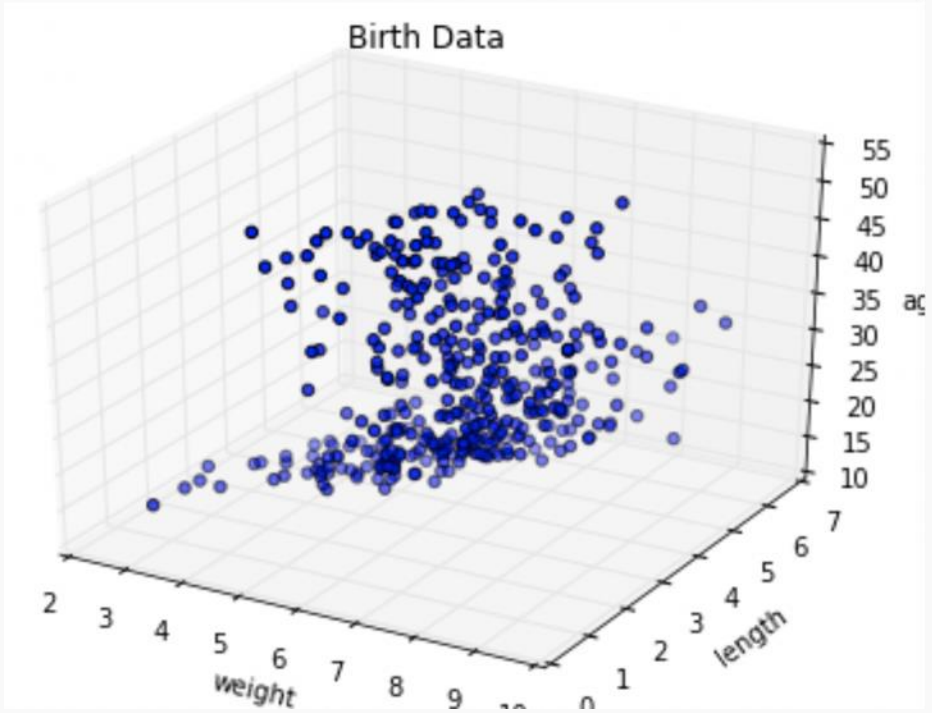


3+ variables

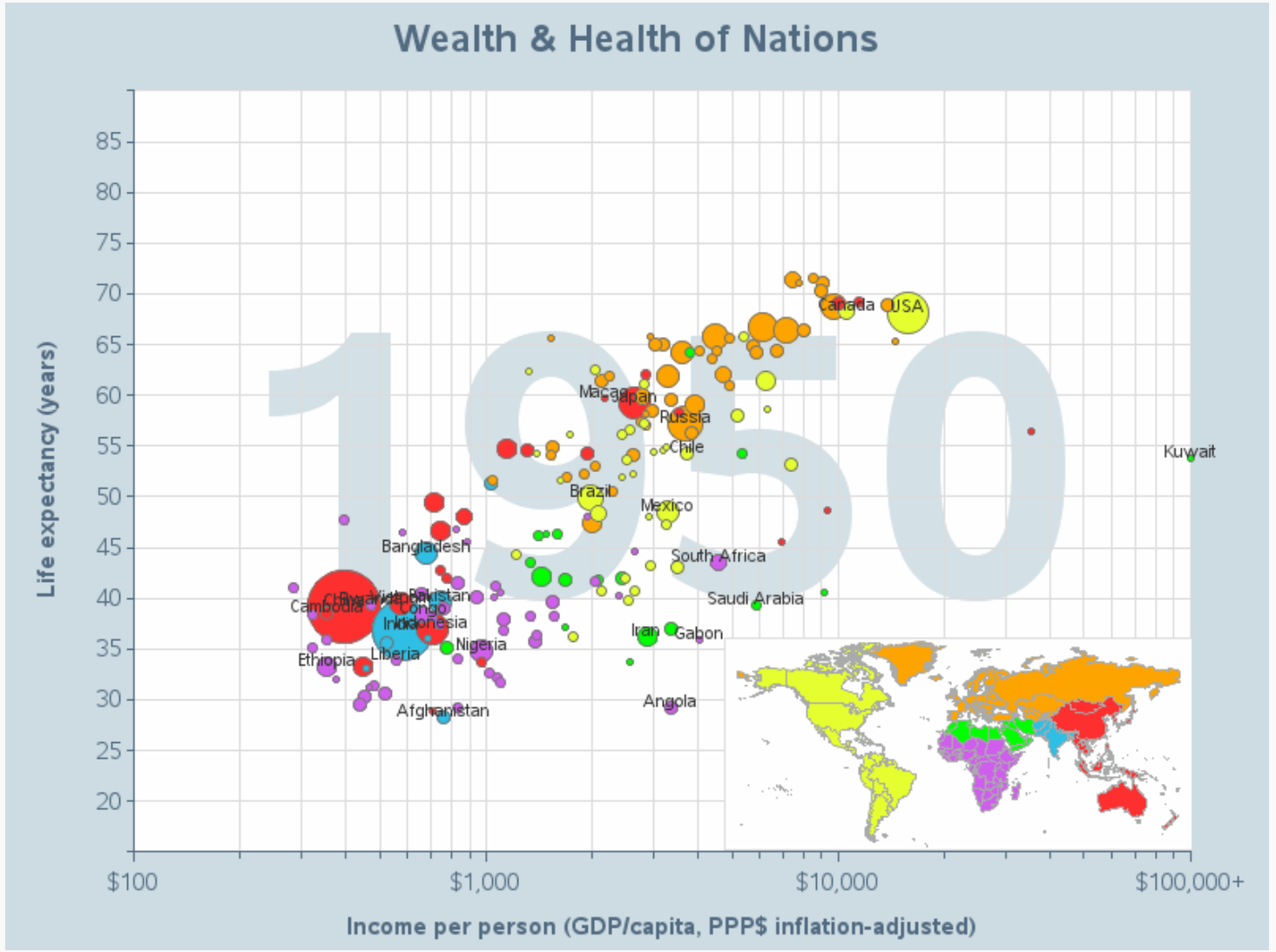
- Visualizing the distribution of 1 variable or the relationship between 2 variables is often straightforward.
- But how can we visualize how 3, 4, or more variables relate?
- It depends on the type of variables (categorical or numeric) that are involved.
- What is the best way to visualize relationships when…
  - all 3 are numeric?
  - 1 or more are categorical?

More dimensions not always better

When the data is high dimensional, a scatter plot of all data attributes can be impossible or unhelpful



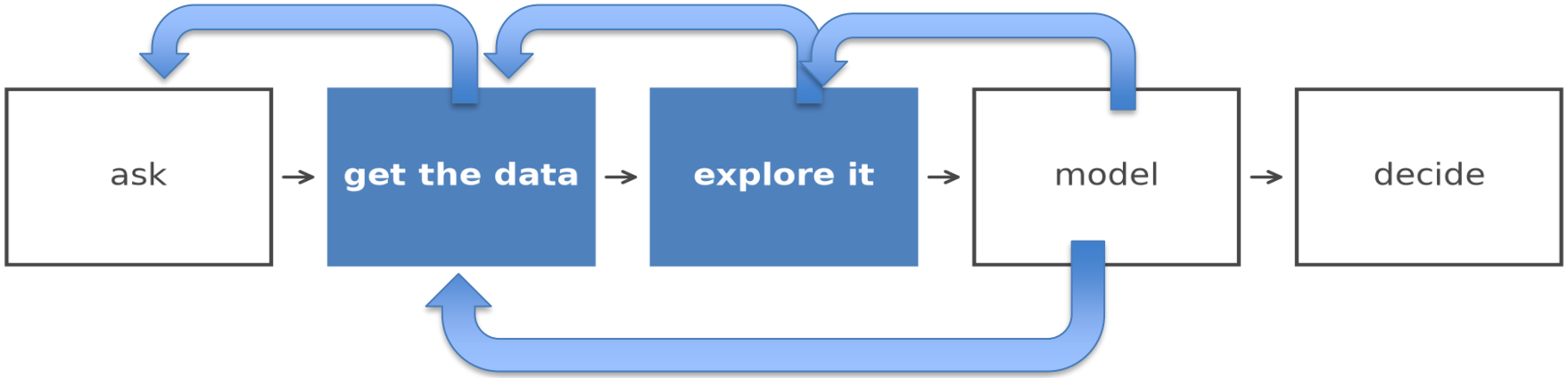
Visualizing 3+ variables



https://www.gapminder.org/tools

Data Science Pipeline

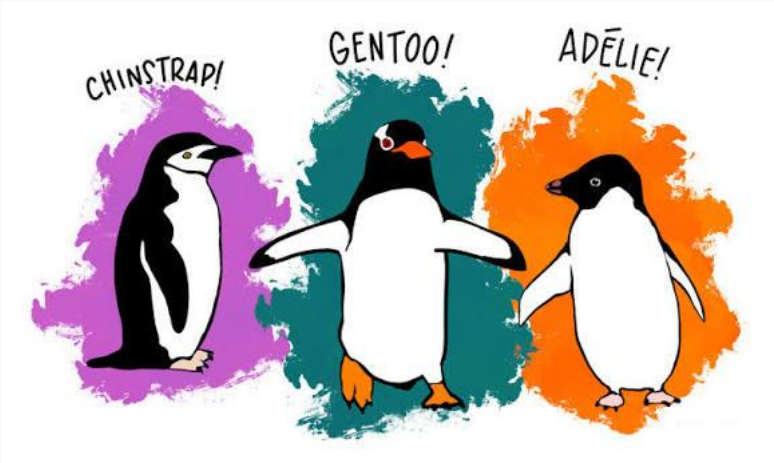
Remember the data science pipeline (from lecture 1):



This process is not linear, and does not always start with asking a question. What if we start with the data?

Exploratory Data Analysis (EDA)

We are given the following dataset. Where should we begin?
How can we generate questions/hypotheses to ask?



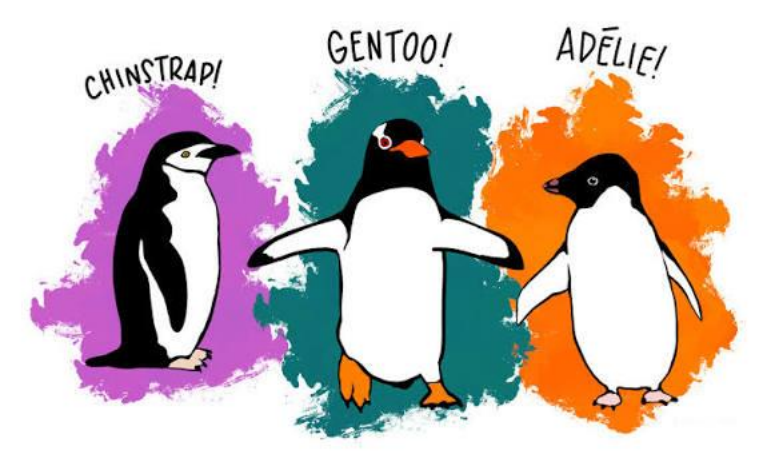

```
penguins = load_penguins()
penguins.head()
```

	species	island	bill_length_mm	...	body_mass_g	sex	year
0	Adelie	Torgersen	39.1	...	3750.0	male	2007
1	Adelie	Torgersen	39.5	...	3800.0	female	2007
2	Adelie	Torgersen	40.3	...	3250.0	female	2007
3	Adelie	Torgersen	NaN	...	NaN	NaN	2007
4	Adelie	Torgersen	36.7	...	3450.0	female	2007

[5 rows x 8 columns]

We should do some exploratory data analysis (EDA) on the data: perform data wrangling, calculate summaries, and create visuals that provide insights into the data and aid in the formation of questions and modeling approaches.

Palmer Penguins -- EDA

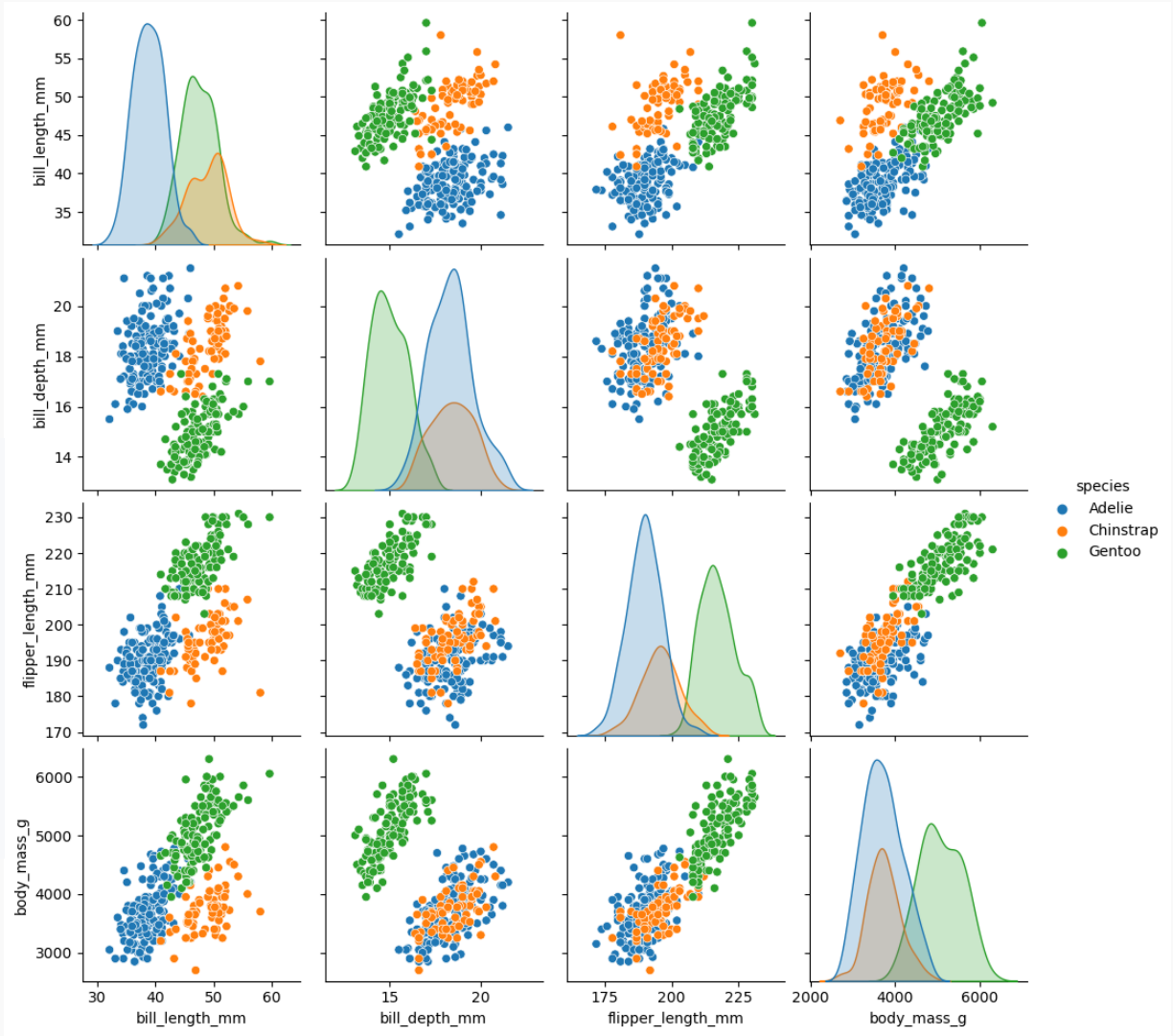



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	species	island	bill_length_mm	...	body_mass_g	sex	year
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3	Adelie	Torgersen	NaN	...	NaN	NaN	2007
4	Adelie	Torgersen	36.7	...	3450.0	female	2007

[5 rows x 8 columns]

What questions arise from the simplistic EDA? What should be our next steps?



sns.pairplot(penguins, hue = " species ")

Lecture Outline: Data, Summaries, and Visuals

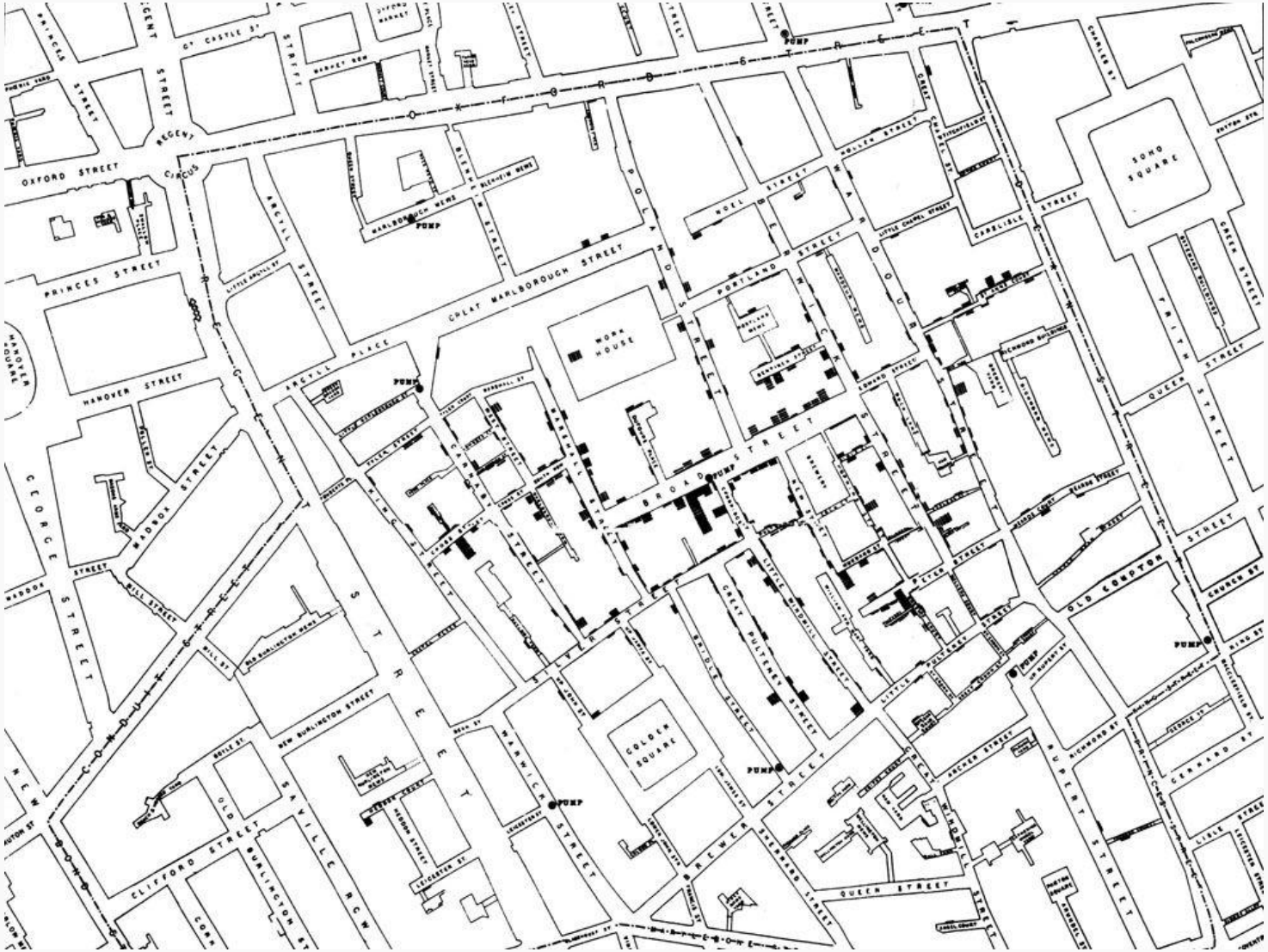
- Exploratory Data Analysis (EDA)
  - Descriptive Statistics
  - Basic Visualizations

- **Historical Interlude**

- Effective Visualizations

Reading: Ch. 1 in *An Introduction to Statistical Learning* (ISL)

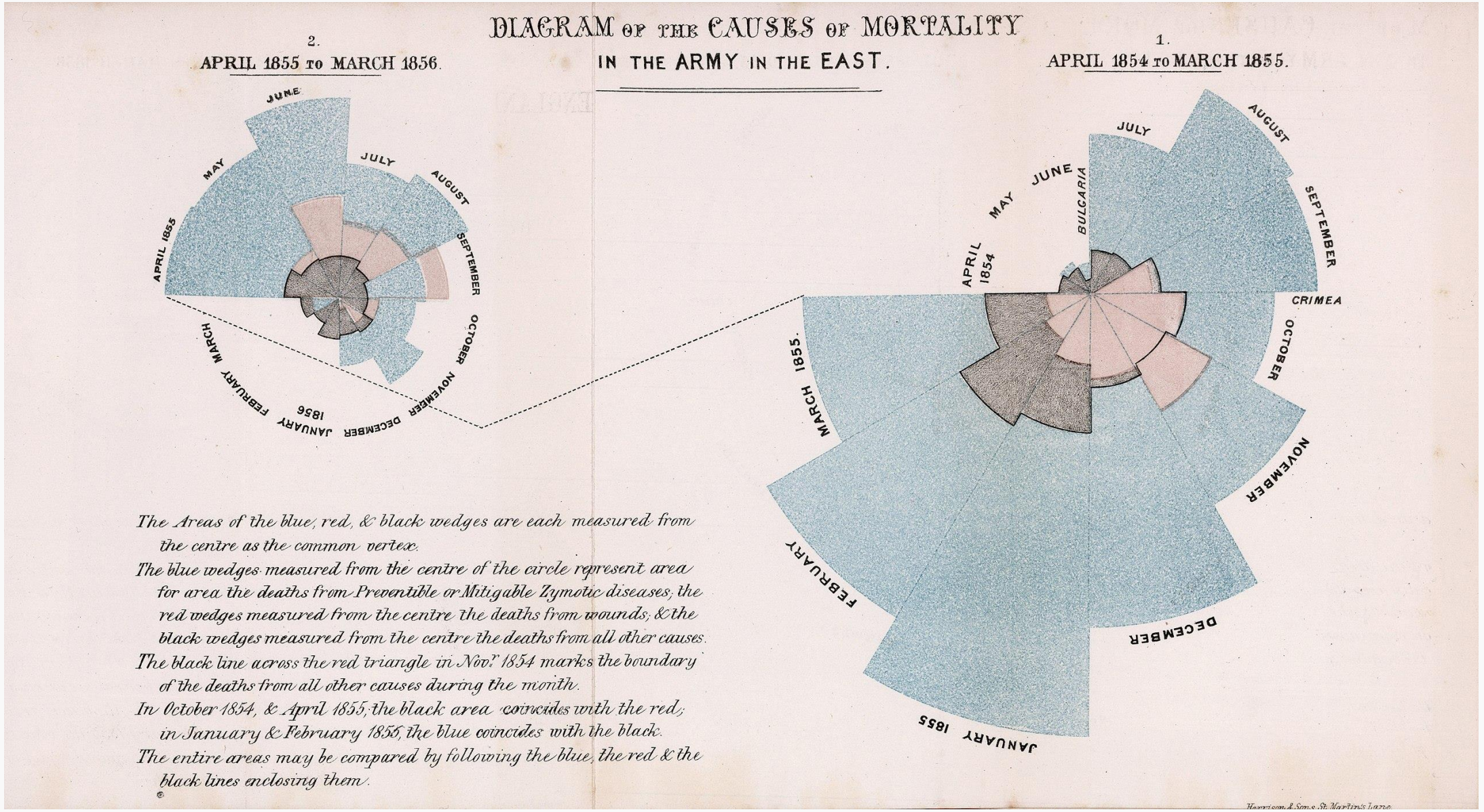
John Snow's Cholera Outbreak (1854)



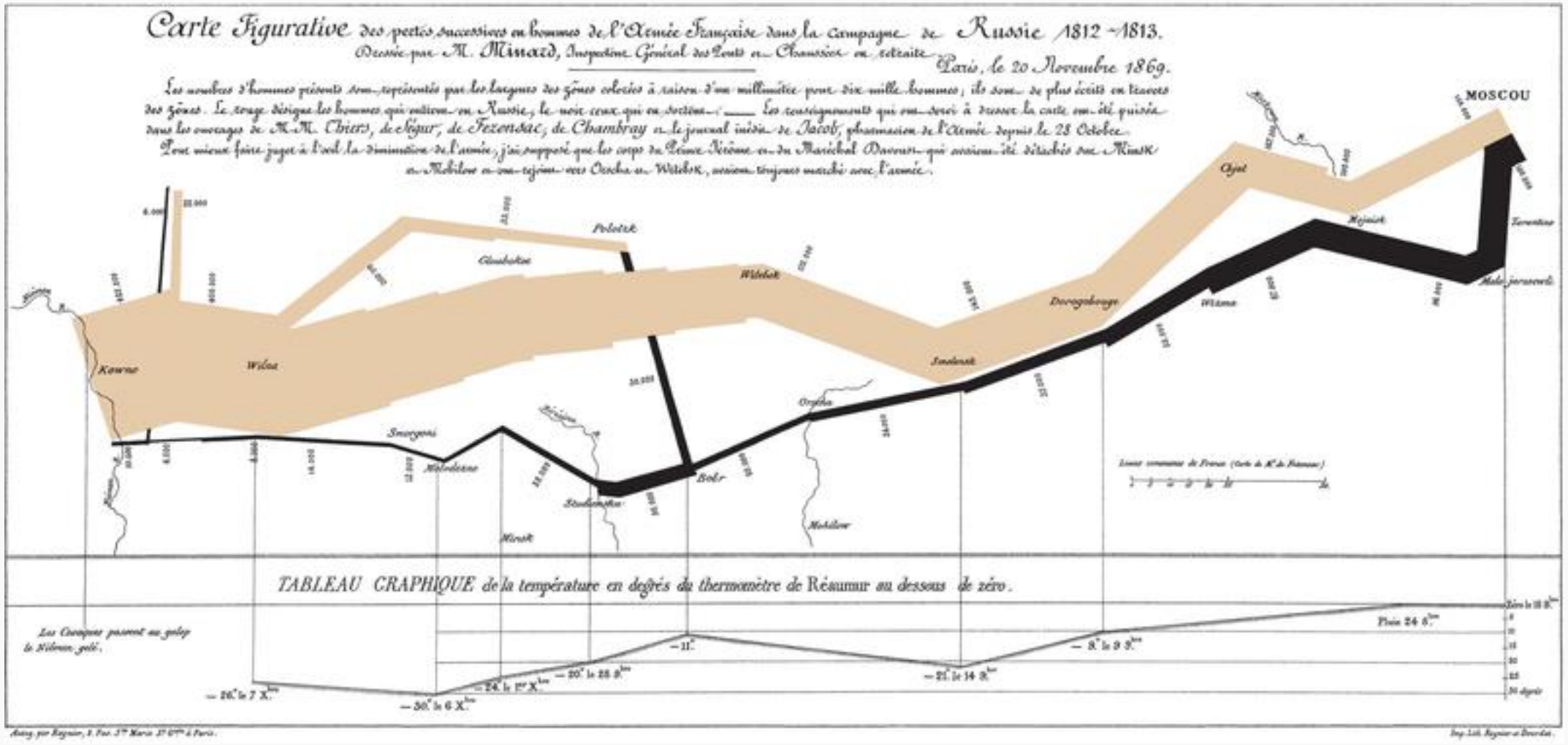


https://en.wikipedia.org/wiki/1854_Broad_Street_cholera_outbreak

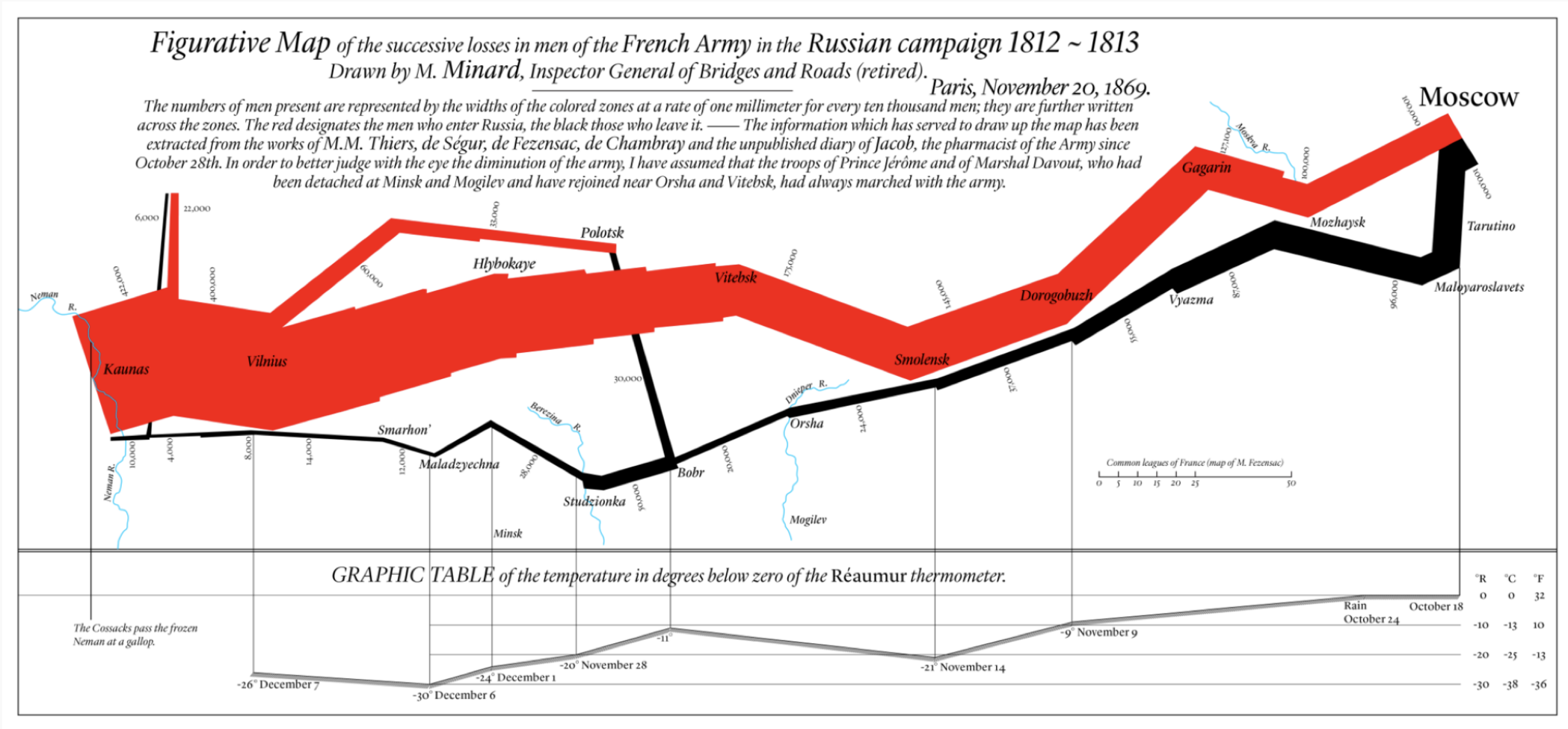
Florence Nightingale's Rose Chart of the Crimean War (1858)



Minard's Visual of Napoleon's March through Russia (1869)



Minard's Visual of Napoleon's March through Russia (1869)



Lecture Outline: Data, Summaries, and Visuals

- Exploratory Data Analysis (EDA)
  - Descriptive Statistics
  - Basic Visualizations

- Historical Interlude

- **Effective Visualizations**

Reading: Ch. 1 in *An Introduction to Statistical Learning* (ISL)

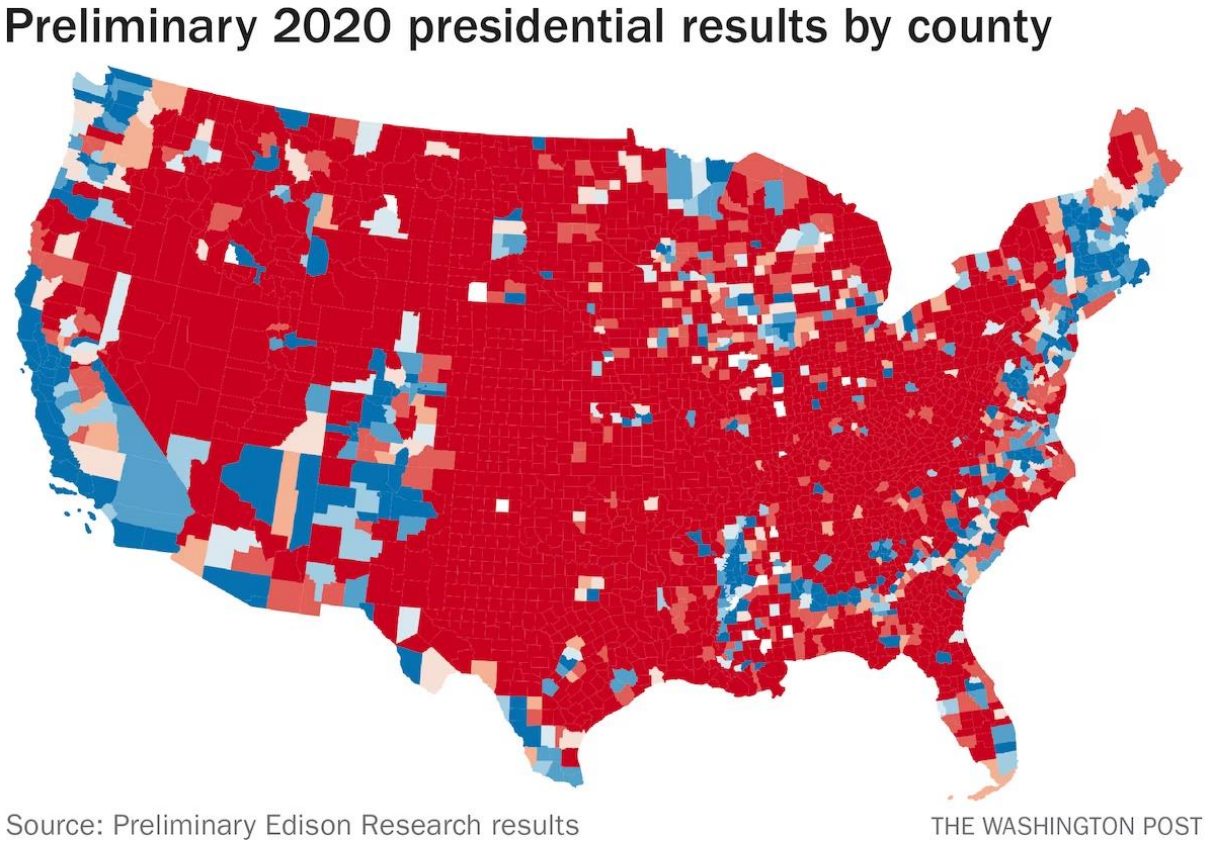
Pointers for Effective Visualizations

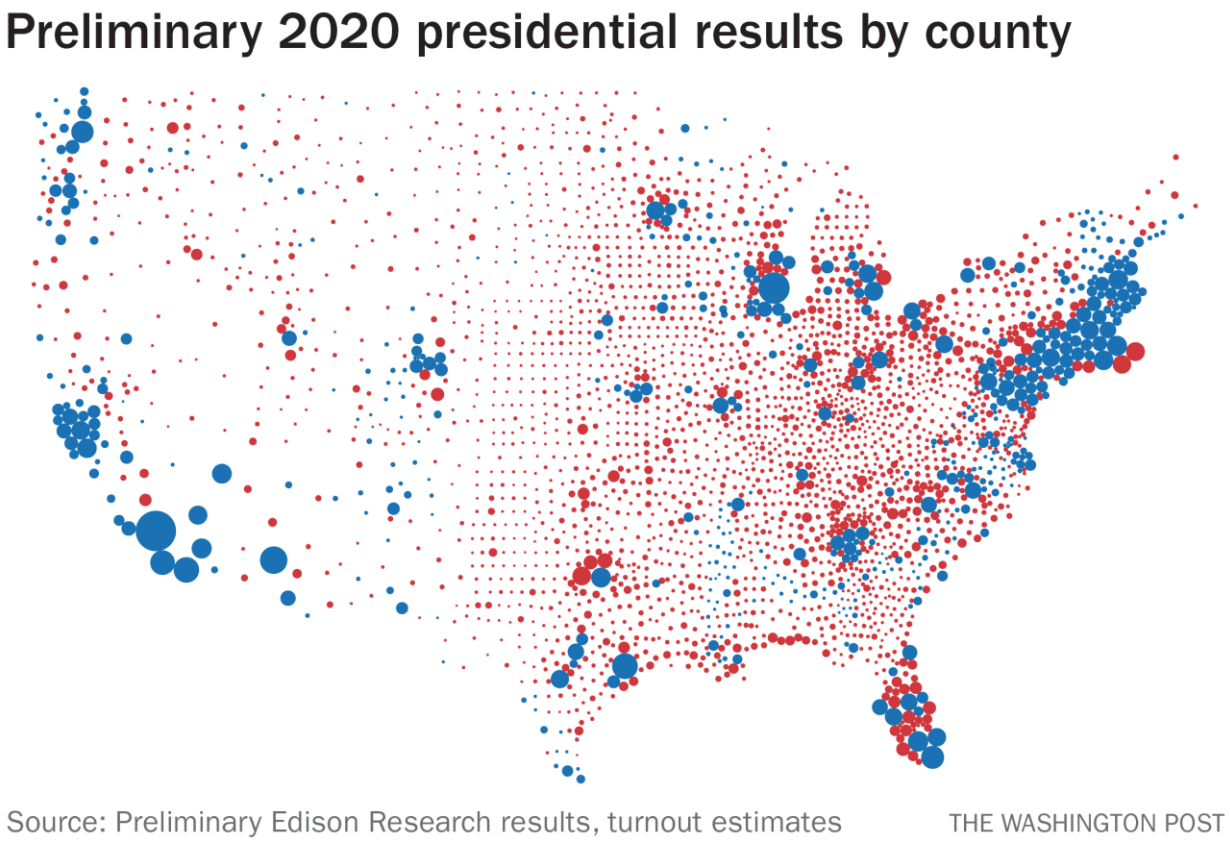
- Have graphical integrity
- Keep it simple
- Use the right display
- Use color strategically
- Know your audience



Graphical Integrity

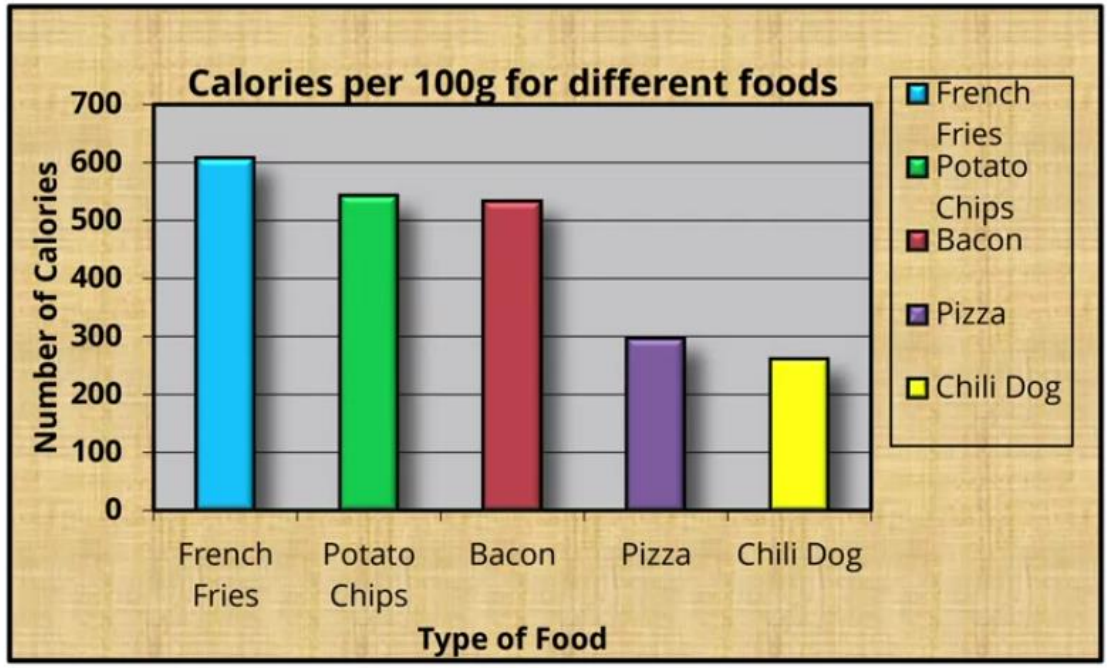
What does this map suggest about the 2020 election?

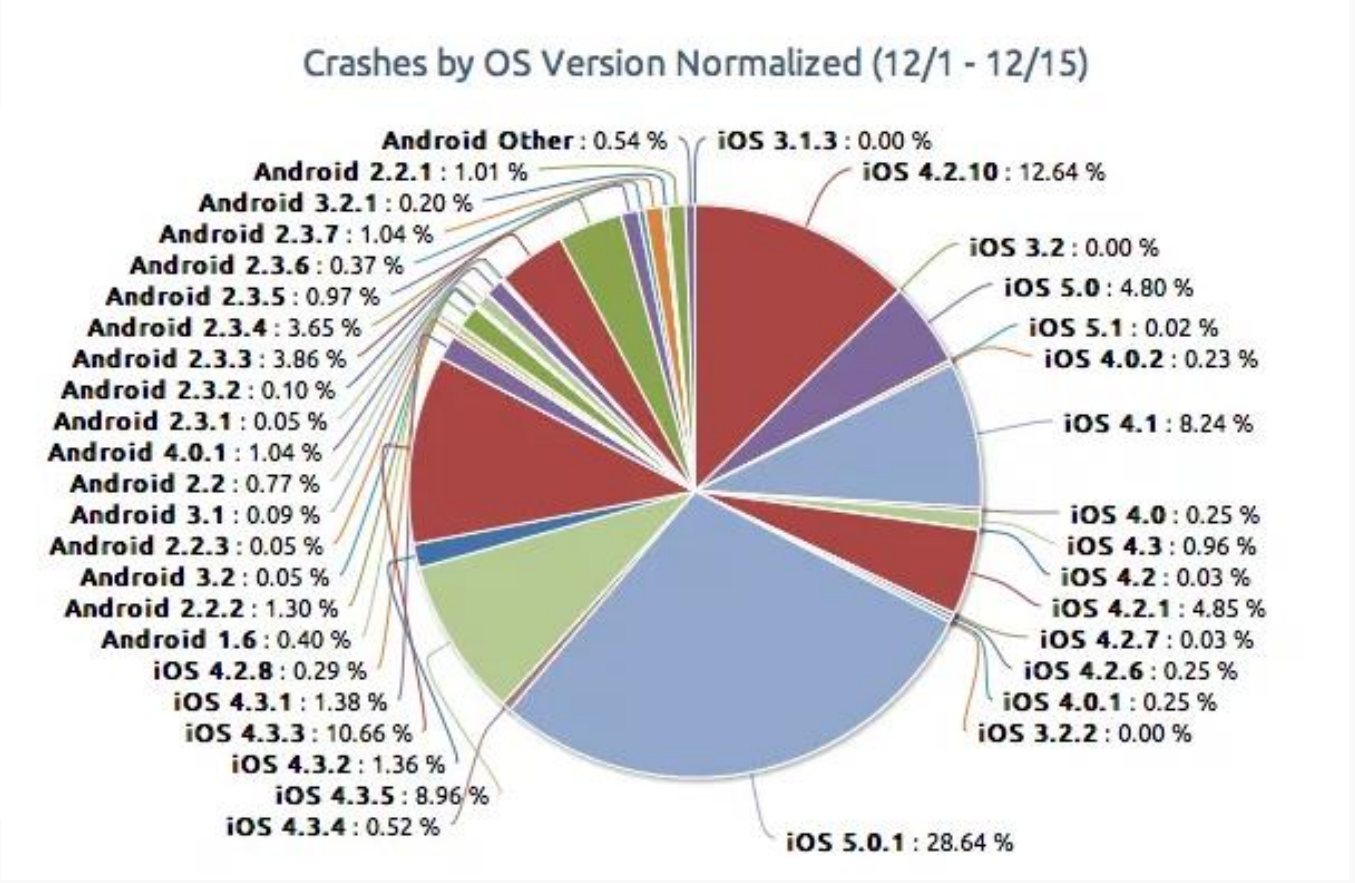




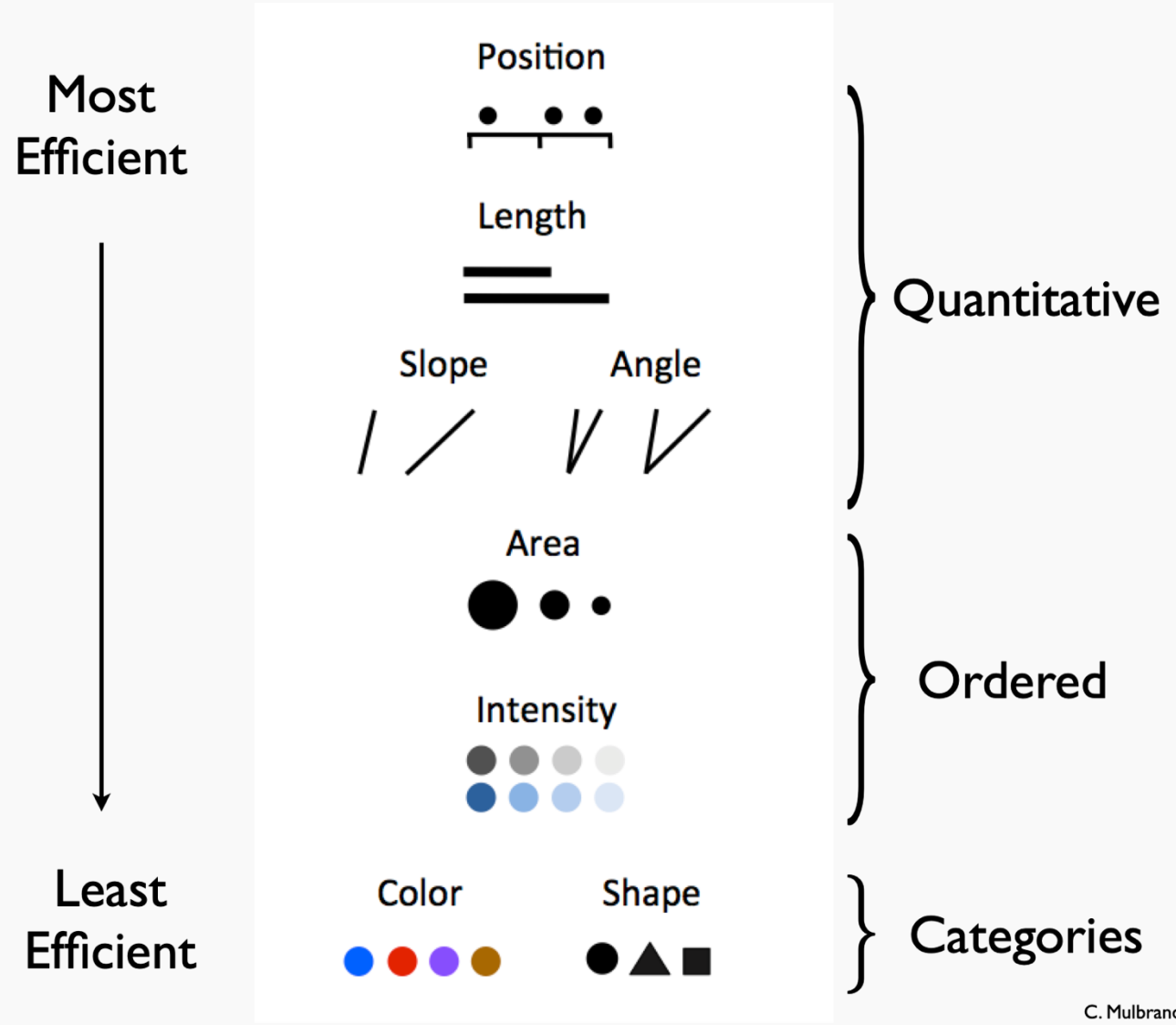
Keep it simple

Avoid Chart Junk





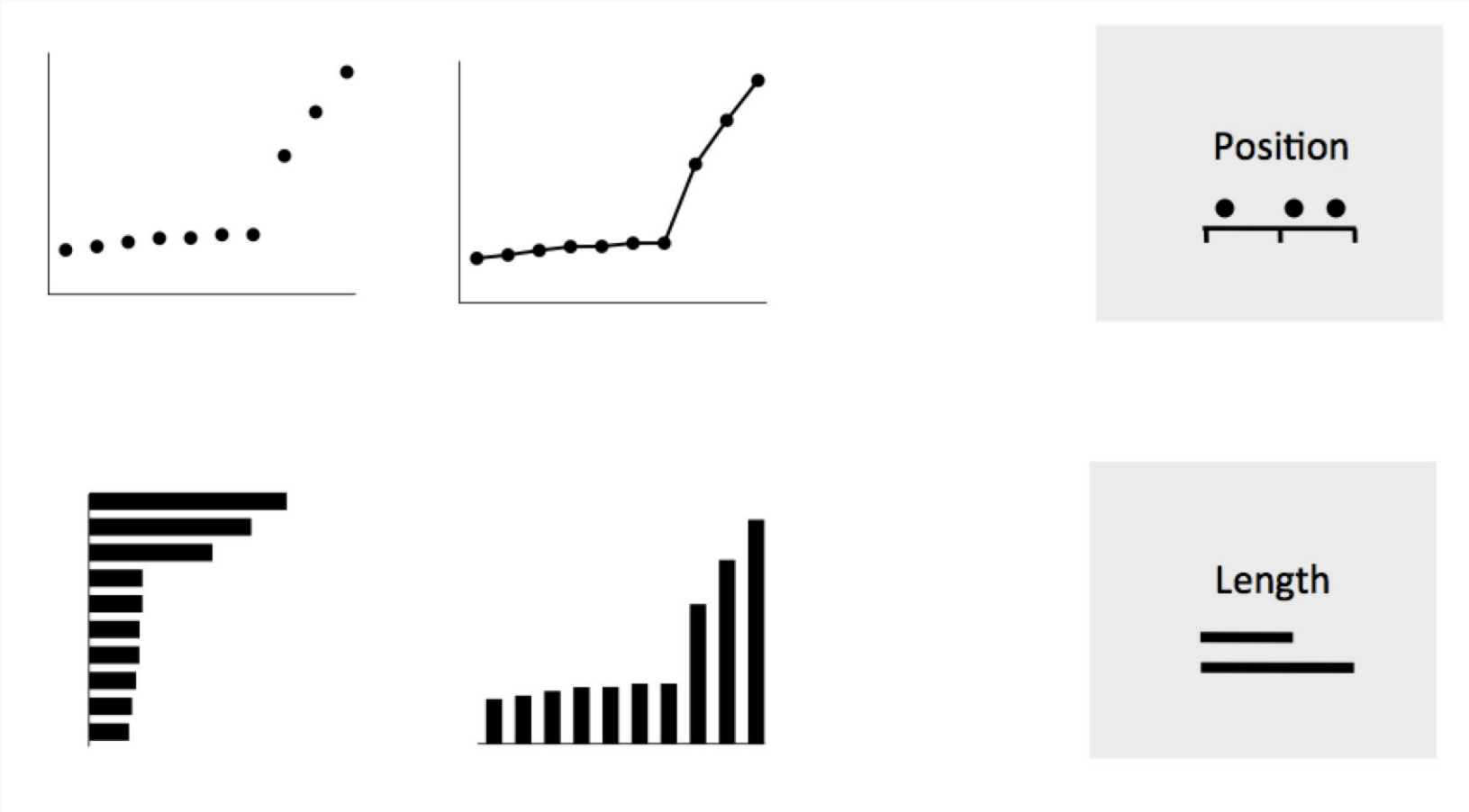
Use the right display



C. Mulbrandon
VisualizingEconomics.com

Use the right display

More effective (numeric scale)



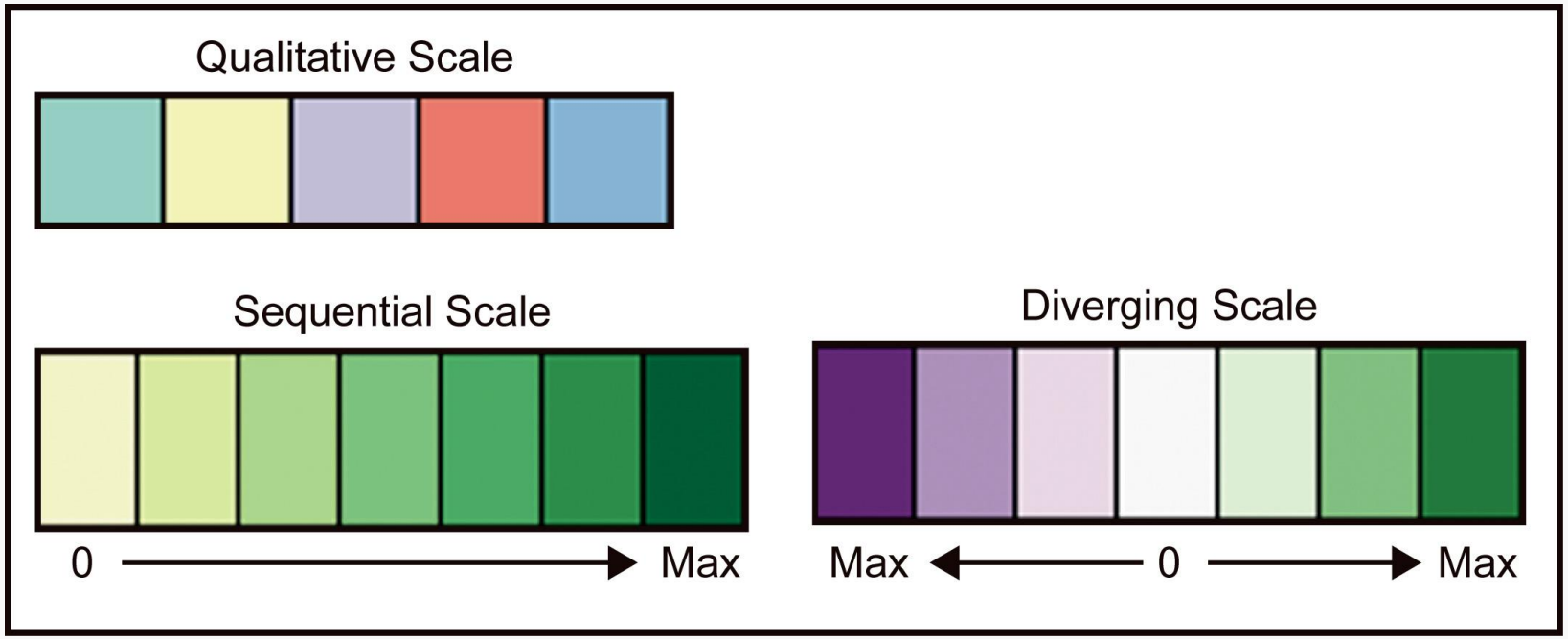
Use the right display

Less effective (numeric scale)



Use color strategically

Type of scale matters:



What are example variables for each?

Use color strategically

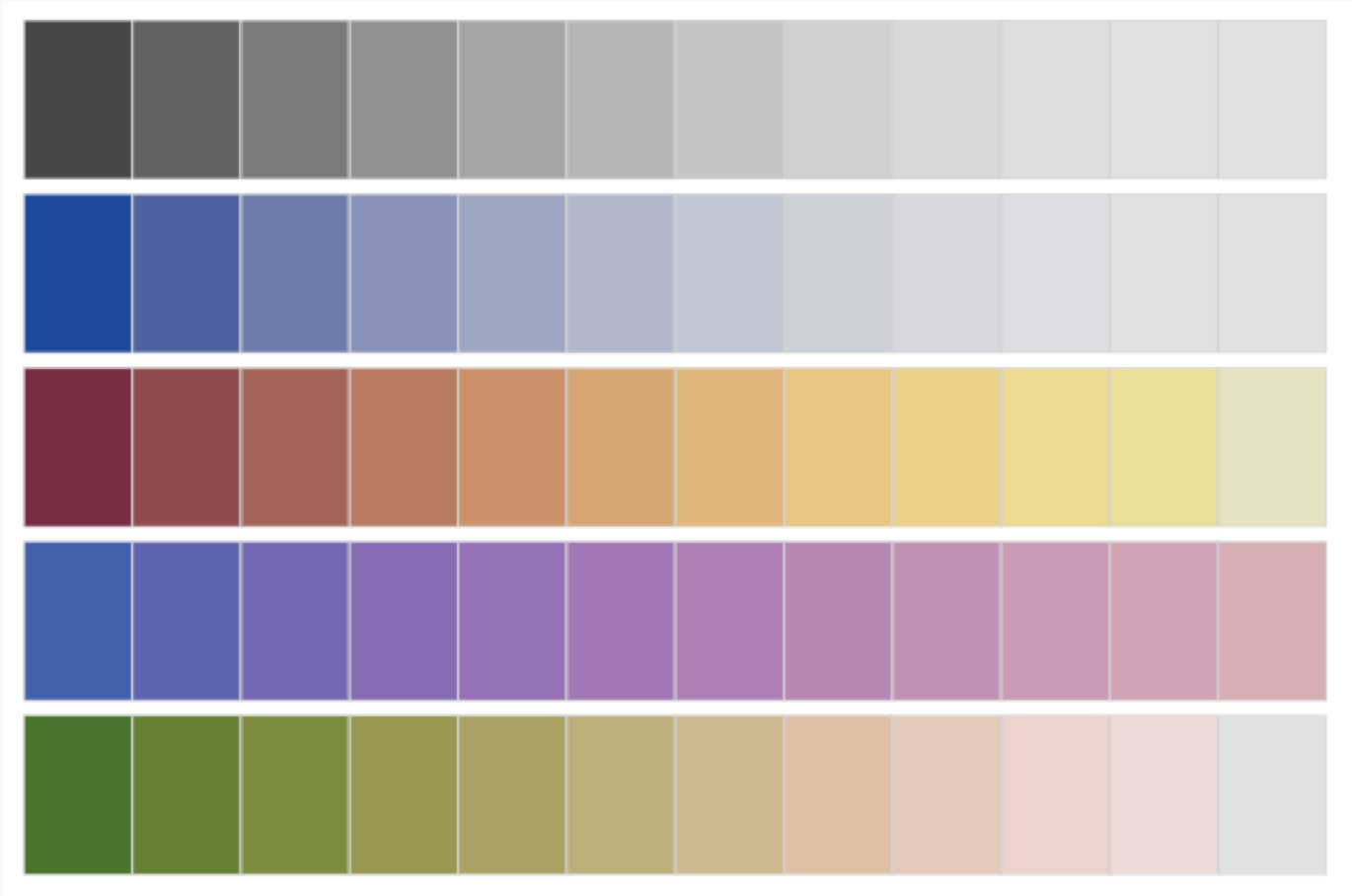
Colors for categories (Do not use more than 5-8 colors at once)





Use color strategically

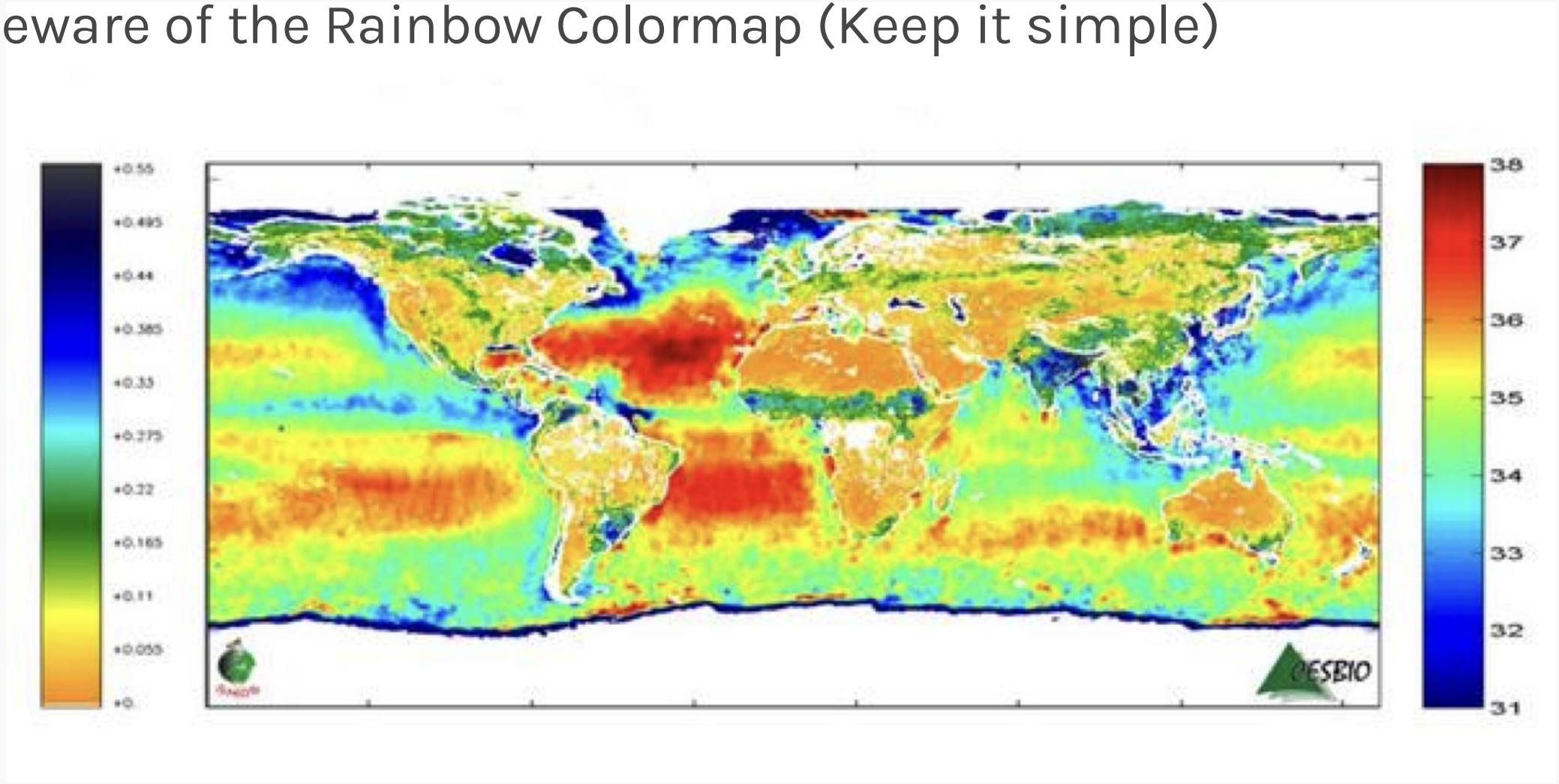
Colors for Ordinal Data -- Vary luminance and saturation



Zeilis et al, 2009, "Escaping RGBland: Selecting Colors for Statistical Graphics"

Use color strategically

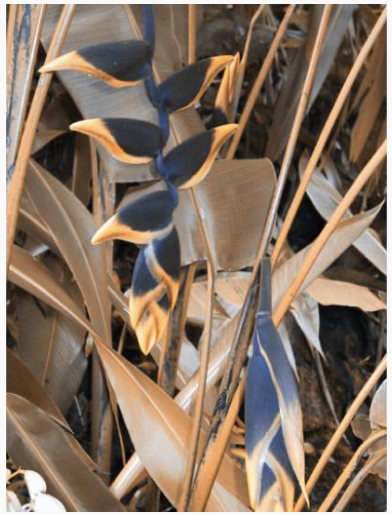
Beware of the Rainbow Colormap (Keep it simple)



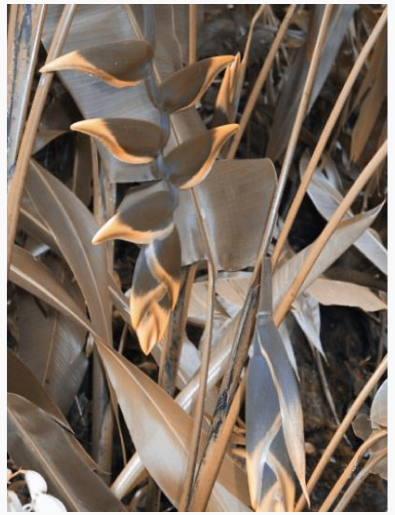
Use color strategically

Be cognizant of color blindness





Protanope



Deuteranope

Red / green deficiencies



Tritanope

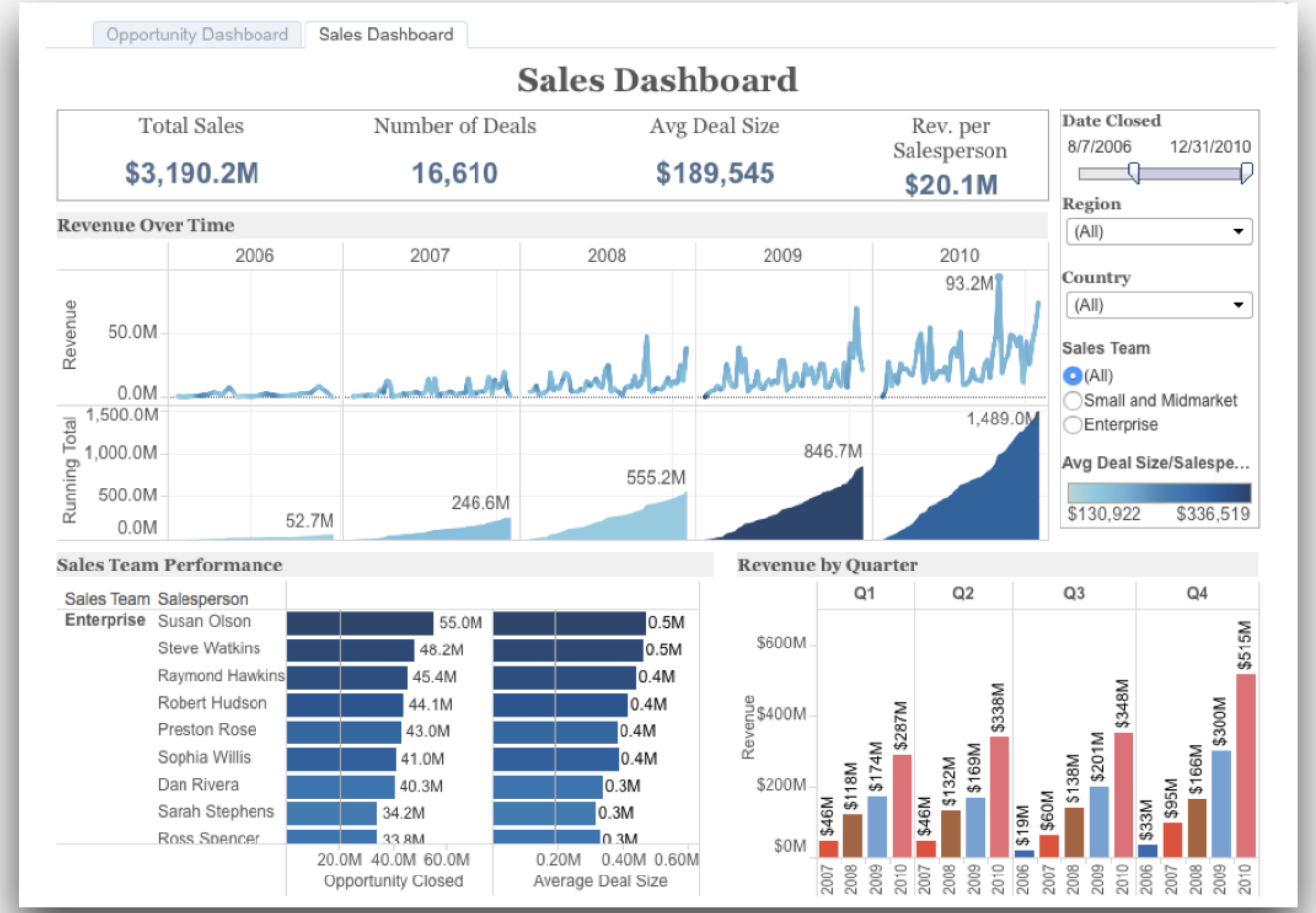
Blue / Yellow deficiency

Know your audience

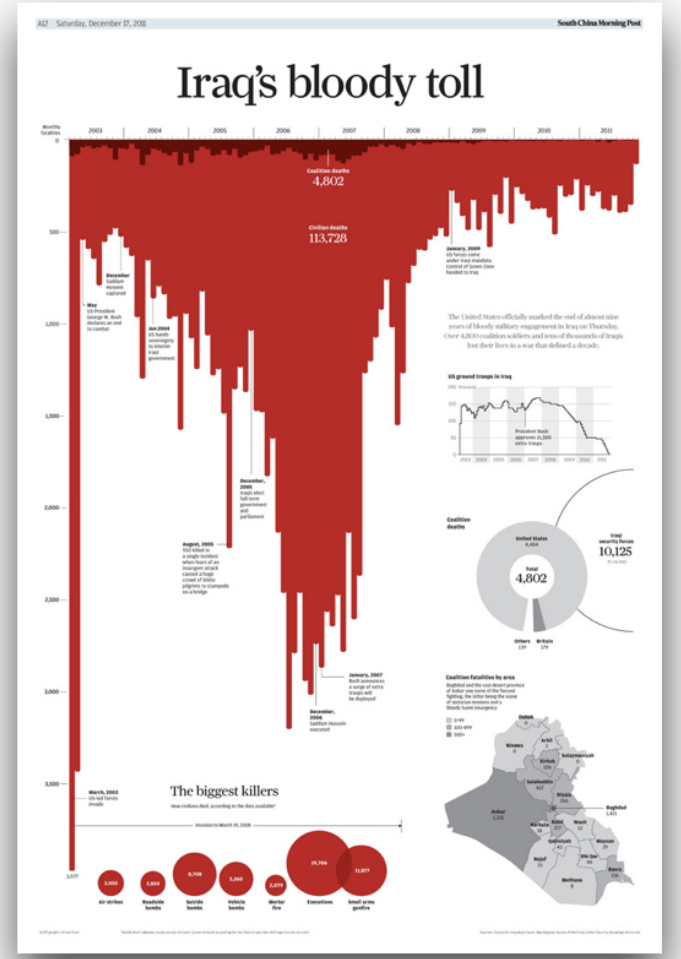
- What do they know?
- What motivates them? What do they desire?
- What experiences do you share? What are common goals?
- What insights can you provide? What tools and “magical gifts”?

Know your audience

Explanatory Neutral

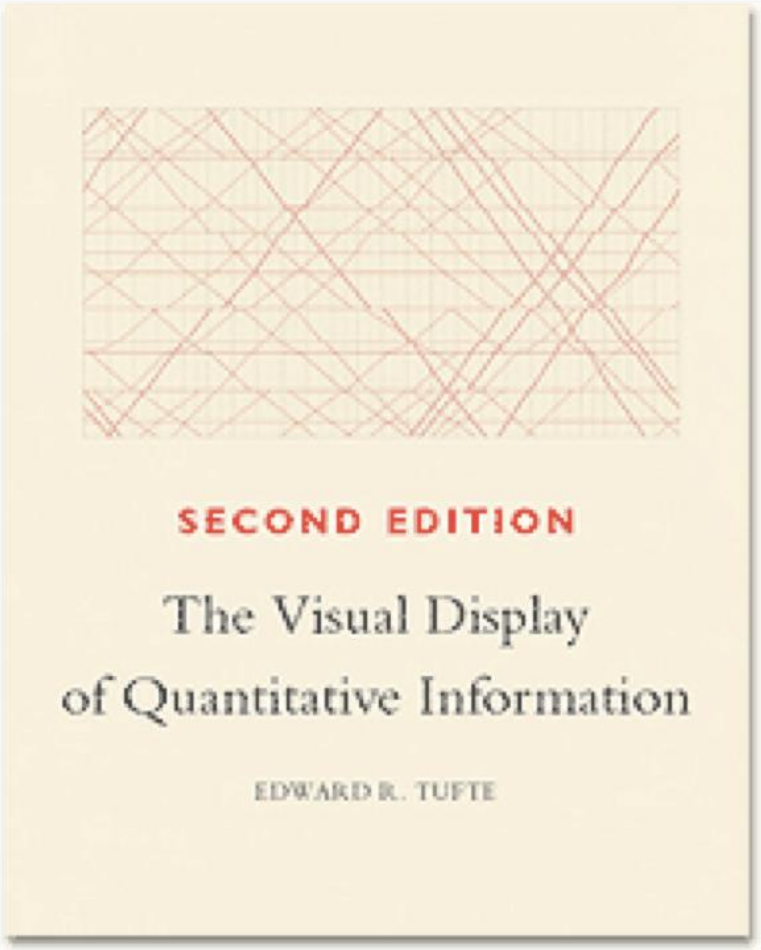


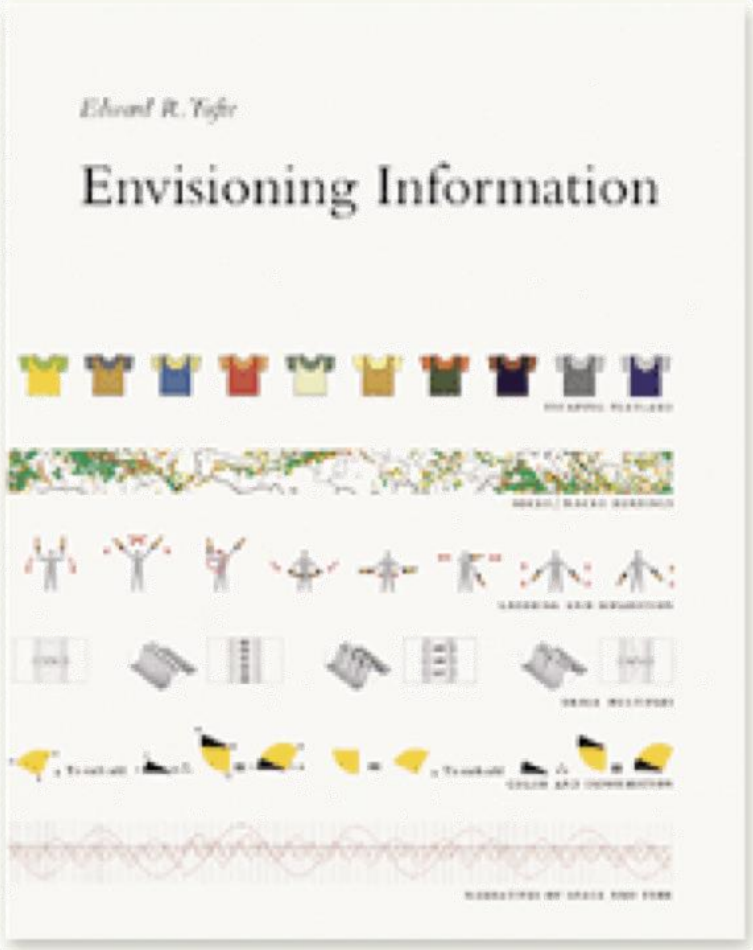
Explanatory Opinionated

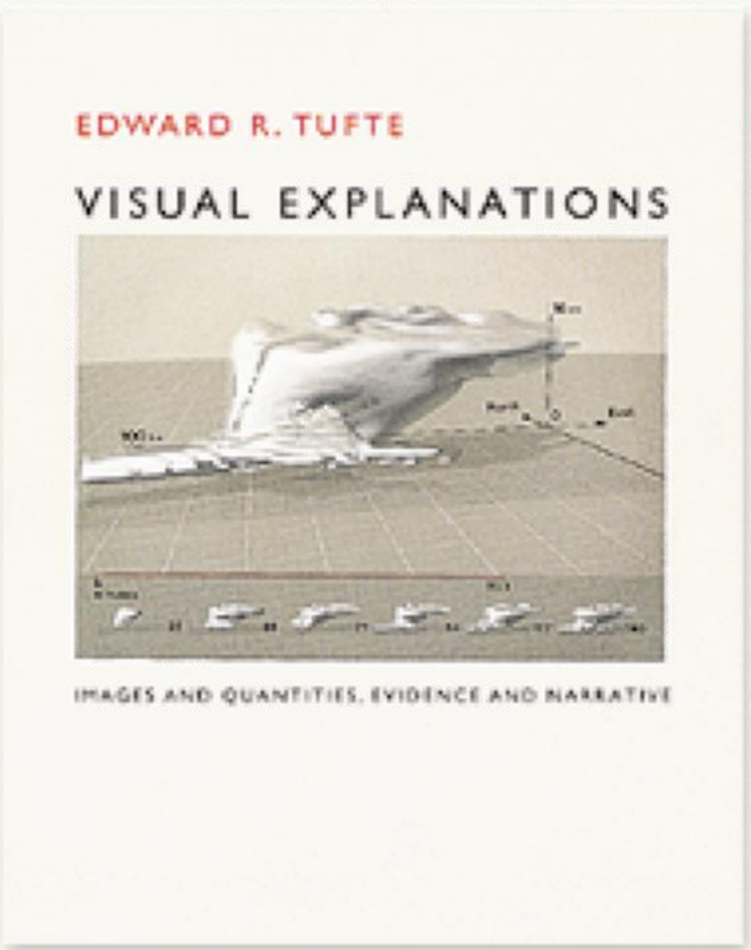


Edward Tufte







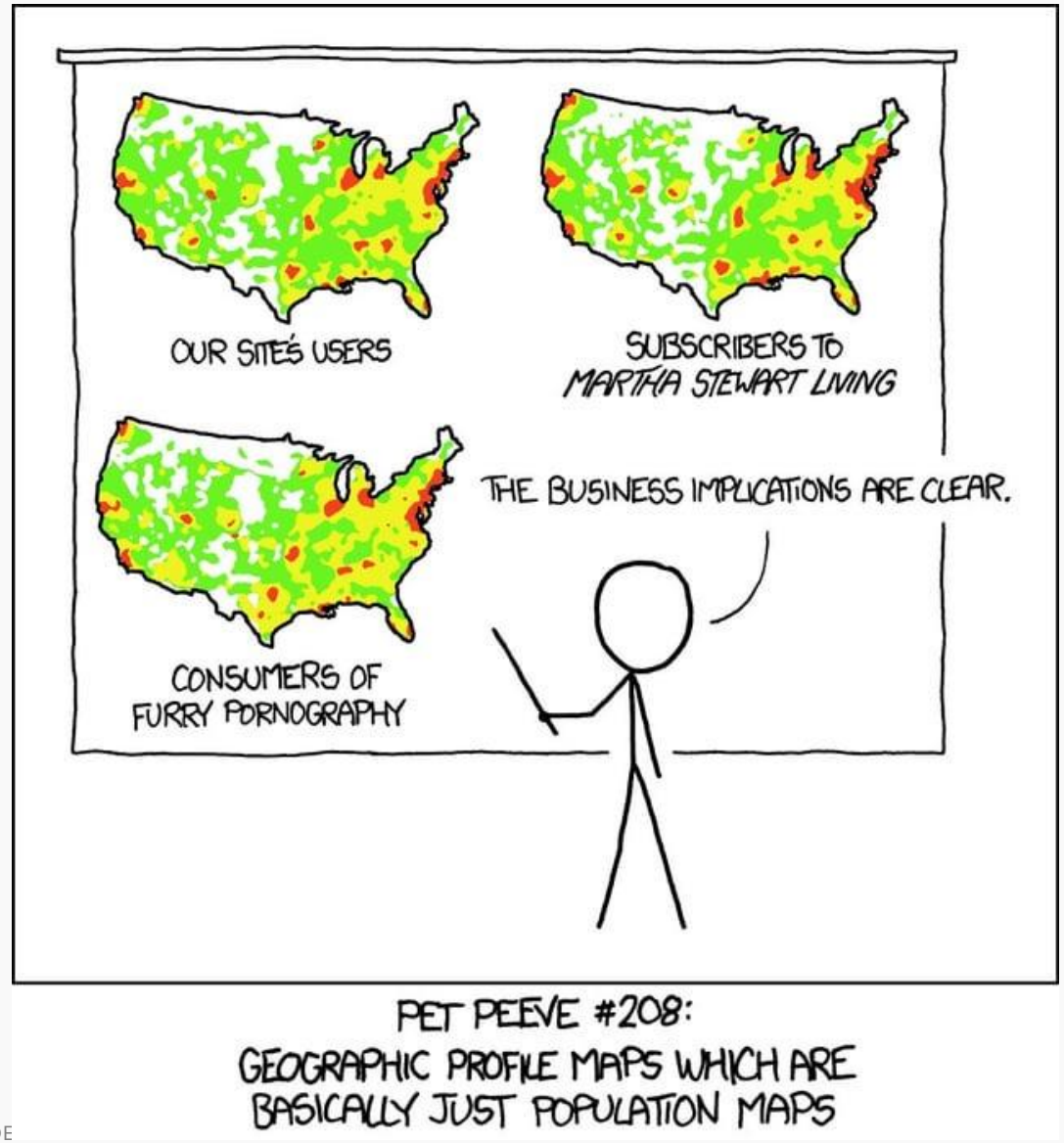


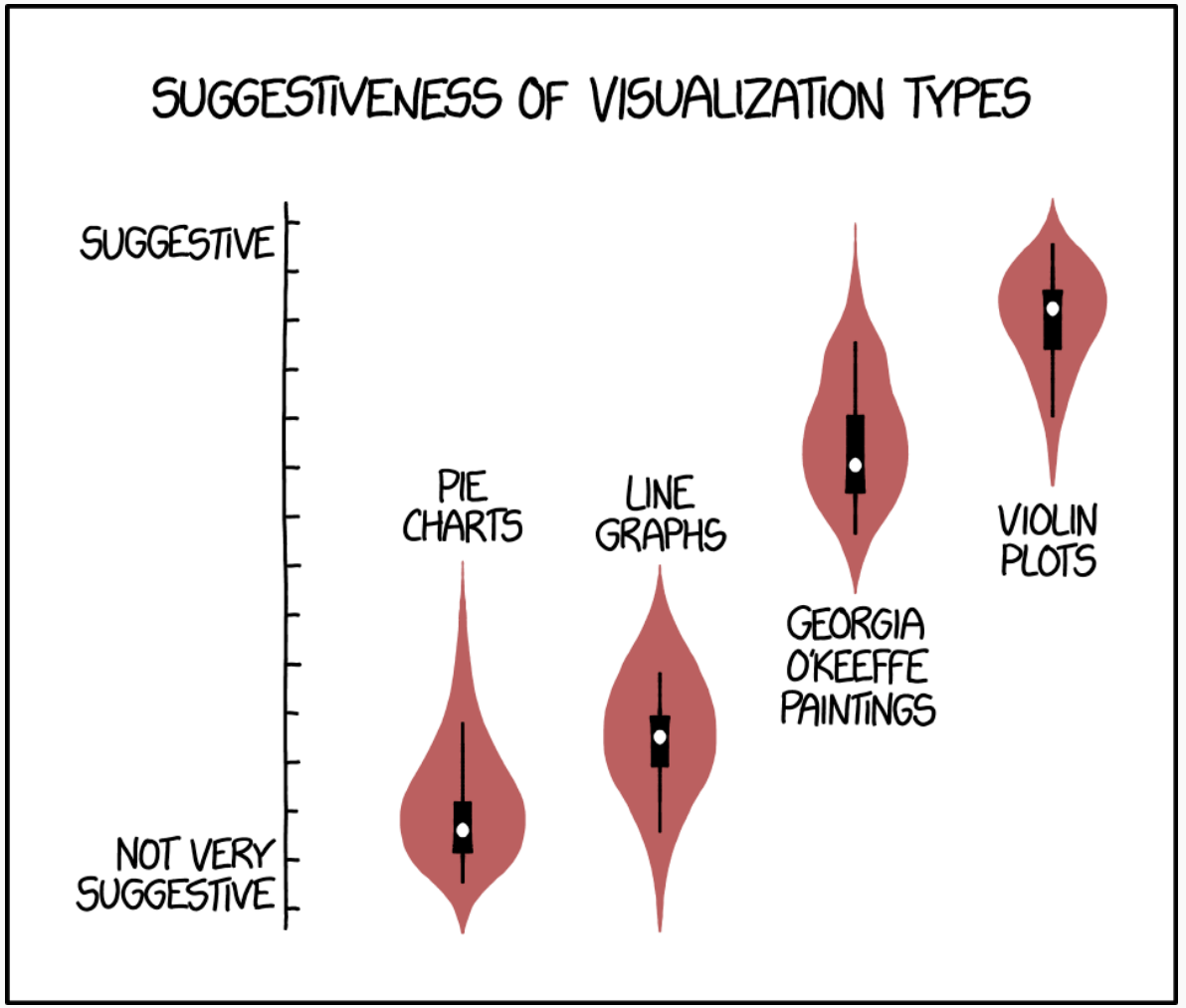
Edward Tufte's Principles of Graphical Excellence

Graphical excellence …

- is the well-designed presentation of interesting data—a matter of substance, of statistics, and of design.
- consists of complex ideas communicated with clarity, precision and efficiency.
- is that which gives to the viewer the greatest number of ideas in the shortest time with the least ink in the smallest space.
- is nearly always multivariate.
- requires telling the truth.

XKCD for the win









CS109A
GAME Time

What is the goal of visualization?

Options

A. To explore the distributions of variables.

B. To explore the data to build hypotheses.

C. To communicate results of your models.

D. To trick and manipulate your audience.

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